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ABSTRACT

With the arrival of the era of big data, the conventional von Neumann architecture is now insufficient owing to its high latency and energy consumption that originate from its separated computing and memory units. Neuromorphic computing, which imitates biological neurons and processes data through parallel procedures between artificial neurons, is now regarded as a promising solution to address these restrictions. Therefore, a device with analog switching for weight update is required to implement neuromorphic computing. Resistive random access memory (RRAM) devices are one of the most promising candidates owing to their fast-switching speed and scalability. RRAM is a non-volatile memory device and operates via resistance changes in its insulating layer. Many RRAM devices exhibiting exceptional performance have been reported. However, these devices only excel in one property. Devices that exhibit excellent performance in all aspects have been rarely proposed. In this Research Update, we summarize five requirements for RRAM devices and discuss the enhancement methods for each aspect. Finally, we suggest directions for the advancement of neuromorphic electronics.

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I. INTRODUCTION

With the rapid growth of information technology, the amount of data processed by computers is increasing. In particular, unstructured data make up ~95% of the processed information.[1] However, processing such complex data with conventional computing is energy-intensive and time-consuming. These limitations mainly originate from the von Neumann architecture, which comprises separated memory and computing units; the exchange between these units during data processing eventually affords considerable impediments in computing power and time (von Neumann bottlenecks).[2,3] For instance, ~40% of power consumption originated from the separated structure of memory and processing units.[4,5] To address this issue, researchers have proposed neuromorphic computing, a method that emulates the behavior of neurons and operations of the human brain, which only requires 25 W of computing power. Since the human brain is composed of parallel connections of neurons, numerous efforts to imitate those structures have been reported by constructing artificial neural network (ANN) structures that enable experience-based computing through the deep learning process of artificial intelligence (AI). ANNs are interconnected groups of nodes, and the learning process comprises repetitions of multiply-accumulate operations.[2,6] Several attempts to implement ANNs in practical devices have been reported.[5–14] Resistive random access memory (RRAM) has remarkable advantages in terms of parallel computing and scalability as it is a simple two-terminal device.[11,15–17] With the advent of various RRAMs, the International Roadmap for Devices and Systems (IRDS) has outlined stringent requirements for fabricating commercial non-volatile memory (NVM) devices in current integrated circuits.[18] Among these requirements are a writing voltage of <3 V, a switching energy of <10 pJ and a time of <10 ns, a writing endurance of ~10^{10} cycles, a high resistance state (HRS)/low resistance state (LRS) ratio of ~10, and a retention time of 10 years at 85 °C with <10% fluctuation. Thanks to recent research efforts, most of these requirements have already been fulfilled and even exceeded.[8,19–22] However, simultaneously satisfying multiple requirements is still challenging.[23,24] Representative example is a trade-off between the

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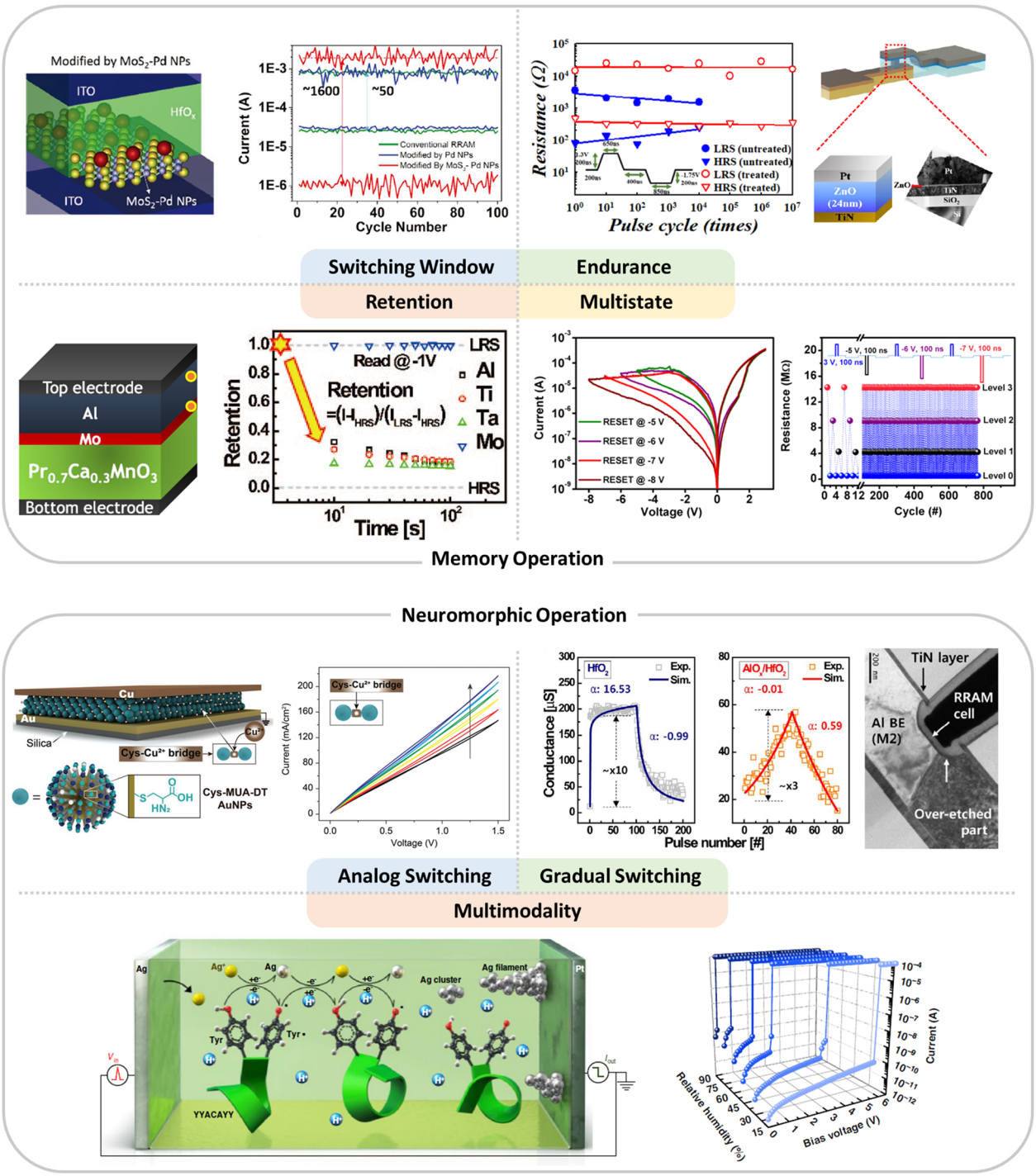


FIG. 1. Five key parameters of RRAM devices: retention, endurance, switching window, synaptic linearity, and multi-modal operation, and various efforts applied to improve each aspect.

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switching window and synaptic linearity of a device. RRAM devices with a high on/off ratio mostly show poor synaptic linearity due to their abrupt resistive switching (RS) behaviors. In this Research Update, five key parameters for RRAM devices have been defined, as shown in Fig. 1,[25,26] and approaches to improve these parameters have been summarized. These methods include material injection (embedding nanostructure in the device or doping new materials in the RS layer/electrode),[8,27–35] additional processes or measurement schemes,[25,36–40] and multimodal device developments[6,38–43] that are not only controlled by electrical inputs but also by other inputs, such as light or humidity. By introducing approaches to improve each memristor feature, we believe that our efforts can contribute on fabricating versatile NVM devices with reliable performance in all aspects. Additionally, this would be an ideal direction for the investigation of neuromorphic electronics.

II. RRAM REQUIREMENTS

A. Switching window

1. Definition and measuring methods

The switching window is the on/off ratio between the low resistance state (LRS) and high resistance state (HRS) of a device, which is measured by simply dividing those two resistance values.[44] From the perspective of neuromorphic device operation, this has rarely been emphasized by researchers due to its trade-off relationship with synaptic linearity. Note that a large dynamic range has the advantages of weight mapping ability and higher accuracy.[45]

2. Switching window improvement through RS layer doping

To achieve a wide switching window, researchers introduced numerous methods, and one of the most effective approaches has been material doping,[31,32,34] which affects the concentration of the filament-forming material in the RS layer. Sedghi *et al.* fabricated a Ta_2O_5-based RRAM device using atomic layer deposition (ALD) and doped the RS layer with the NH_4OH solution, which induced a wide switching window, improved the stability, and lowered the reset current.[32] The undoped device shows unstable behavior at both the HRS and LRS with a narrower switching window. This phenomenon has been explained using density functional simulations and density of state calculations. According to those, N dopants substitute O atoms adjacent to oxygen vacancies, eliminating additional conductive paths.

Additionally, the RS behavior of a Cr-doped ZnO-based RRAM device has been reported by Li and Su.[34] Two kinds of devices were fabricated to investigate the effects of Cr doping. Undoped and Cr-doped ZnO (2 wt. % Cr) were deposited on indium zinc oxide (IZO) substrates at 800 °C using a co-sputtering system. The on/off current ratios of the undoped and doped devices were 3.51×10^1 and 9.12×10^2, respectively. Through x-ray photoelectron spectroscopy depth-profiling analysis, these phenomena have been attributed to oxygen vacancy concentration differences.

3. Switching window improvement through top electrode doping

Engineering not just the RS layer but also the top metal expands the memory window [Figs. 2(a)–2(c)].[46] By adopting an

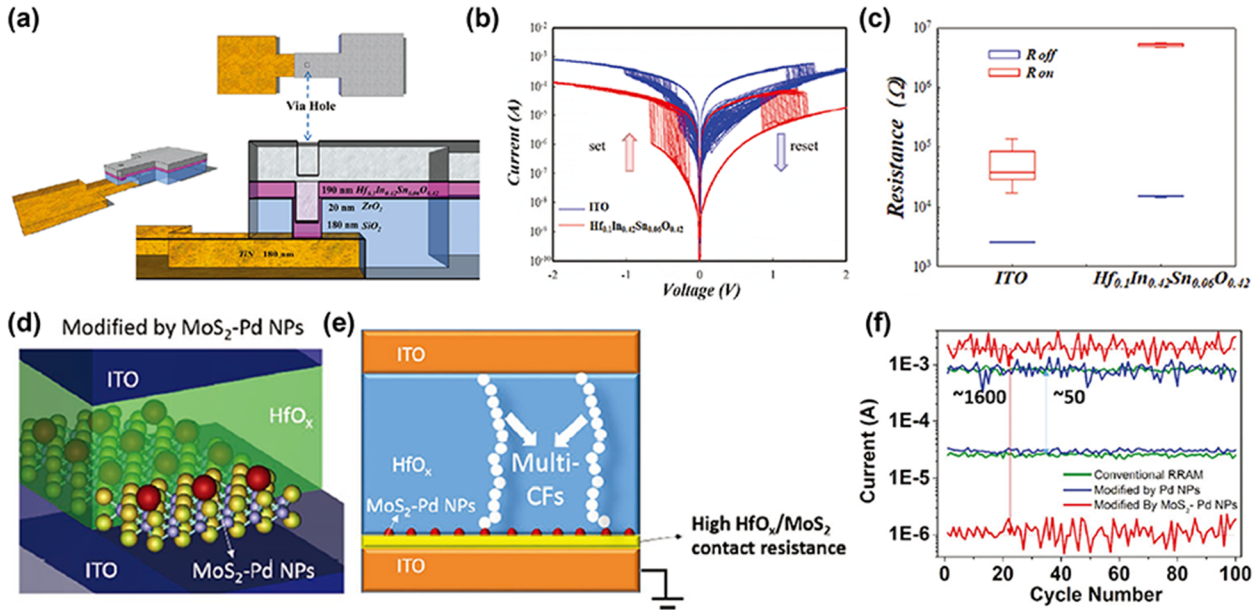


FIG. 2. (a) The 3D view and section view of the top metal doped $Hf_{0.1}In_{0.42}Sn_{0.06O0.42}/ZrO_2$/TiN device structure and (b) I–V curve of the device. (c) The on/off state resistance ratio comparison of the top metal doped device with the original one. (d) 3D sketch of ITO/HfO_x/MoS_2–Pd NPs/ITO stacked RRAM and (e) CF forming behavior. (f) LRS and HRS current value of conventional ITO/HfO_x/ITO RRAM (green), ITO/HfO_x/Pd NPs/ITO RRAM (blue), and ITO/HfO_x/MoS_2–Pd NPs/ITO RRAM at −0.1 V of RRAM reading voltage. (a)–(c) Copyright 2020, ©The Royal Society of Chemistry. (d)–(f) Copyright 2017, Wiley-VCH.

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Hf–In–Sn–O composite instead of indium tin oxide (ITO) as the top layer [Fig. 2(a)], the working parameters (low power consumption, working stability, and memory window) of the RS device have been effectively enhanced [Figs. 2(b) and 2(c)]. Particularly, the average LRS and HRS resistances of the top-metal-doped device are 20 kΩ and 6 MΩ, respectively, while those of the undoped device are 2 kΩ and 40 kΩ, respectively; the memory window was enlarged 15 times. In the undoped device, electron carriers easily find conduction paths in the top electrode as ITO conducts via the oxygen vacancy mechanism. On the contrary, in the top-metal-doped device, a semiconductor-like layer is formed due to accumulation around Hf atoms (by electron clouds and oxygen ions). These findings are consistent with the difference in the on/off ratio of the devices, as shown in Fig. 2(c), which indicates that the LRS current of the top-metal-doped device was lower than that of the undoped device.

4. Switching window improvement through nanostructure embedding

Embedding nanostructures into the RRAM device can be a solution to widen the switching window.[33,35,39] Wang *et al.* introduced MoS_2–Pd nanoparticles (NPs) into HfO_x and constructed an ITO/HfO_x/MoS_2–PdNPs/ITO stacked RRAM device [Figs. 2(d)–2(f)].[33] Owing to the large contact resistance from the MoS_2/HfO_x interface, the resistance of the HRS of the MoS_2-embedded device is more than an order of magnitude larger than that of the reference device. Moreover, the NPs enlarged the switching window by 32 times that of the reference device.

Wang *et al.* introduced metal nanoisland (NI) arrays in the middle of an RS HfO_x layer.[35] Pt, Ti, and Ag NI arrays were embedded, and each device showed enhanced on/off ratios, cycle-to-cycle resistance uniformity, and low operating voltages. Electrostatic force microscopy and conductive-atomic force microscopy measurements revealed that the enhanced RS behavior originates from the local confinement of the electric field, inducing preferential oxygen reduction between the top electrode and NI array.

Sun *et al.* fabricated an Au/ZnO nanorod/AZO sandwich-structured memristor and developed a rapid hydrogen annealing treatment.[39] With the annealing process, three orders of on/off ratio enhancement were observed. This improvement stemmed from the high-concentration oxygen vacancy layer and improved the crystallinity resulting from the hydrogen annealing.

5. Insights into the switching window

The switching window—the resistance ratio before and after a set process—is typically measured by comparing the current values of the HRS and LRS at a specific voltage (usually 0.1 V). Devices with a narrow switching window have an insufficient number of states during weight modulation, significantly lowering their recognition accuracy. Since the early 2010s, the switching window has been an important indicator in memory applications;[47,48] thus, various attempts to improve the window through annealing[39] and doping[31,32,34,46] have been reported. Recently, switching window enhancements through material modulation have been achieved through approaches such as nanocomposite embedding[33,35,39] and selective etching.[40]

B. Endurance

1. Definition and measuring methods

The endurance of RRAM devices refers to the number of times RS occurs under a sufficient HRS and LRS ratio.[44,49,50] Long-term weight update is required to construct a device that operates as a neuromorphic chip. This requirement necessitates good endurance characteristics. Numerous methods for evaluating endurance characteristics have been already reported, and the most simple method among them would be repetition of current–voltage sweep, in which R_{LRS} and R_{HRS} are extracted at a specific read voltage (typically 0.1 V).[51] This method is simple and reliable because it provides the correct RS of the device in each cycle. However, it often failed to accurately measure the endurance characteristics due to the long measuring time and is thus affected by the retention ability. Therefore, researchers developed constant pulsed voltage stress (PVS) schemes by measuring the driven currents.[51,52] The PVS schemes consist of two types of voltage pulses: one pulse with a relatively large Vwrite used to set/reset the device and a succeeding pulse to read the resistance of an RS cell after each voltage stress. However, Lanza *et al.* argued that the endurance characterization method must contain current measurement in every cycle, as, otherwise, the local RS failure could be missed, leading to an endurance overestimation.[53] According to IRDS 2021, the required endurance for a practical neuromorphic device is $>10^{10}$ cycles,[18] which means that an unrealistic measuring time (more than a year) is required. Therefore, measuring the endurance with a current-visible method until certain cycles (10^6–10^7) for multiple devices to confirm acceptable RS and then applying additional endurance measurement with only few current read pulses, which were randomly selected for faster measurement, would be practical way to measure endurance of the devices.

2. Endurance of current devices

In fact, extraordinary endurance characteristics have already been reported by Lee *et al.* in 2011.[22] The researchers deposited a bilayer of a TaO_{2-x} base and a few-nm-thick Ta_2O_{5-x} layer using reactive sputtering [Fig. 3(a)]. The resistance of the insulating Ta_2O_{5-x} layer (upper layer) was kept at 10^8–10^9 Ω, while the resistance of the TaO_{2-x} base layer was maintained at 10^3–10^4 Ω to control the programming current and the device stability. Endurance characteristic of the fabricated device was measured under the voltages of V_{SET} = −4.5 V, V_{RESET} = 6 V, and V_{Read} = −0.5 V, and the extraordinary value (more than 10^{10} cycles) at every oxygen partial pressure condition (2.1%, 2.7%, and 3.0%) has been confirmed. Notably, the highest endurance value of 10^{12} cycles was observed from a 30 × 30 μm^2 cell with a base layer oxygen partial pressure of 3% [Fig. 3(b)].

3. Endurance failure mechanisms

RRAM structured devices are mainly operated by applying voltage bias to metal electrodes to form CFs in the RS layer. Since RS-based devices have different electrical values after each cycle, it is impossible to predict the exact movement of each atom inside the device, and it should be avoided to do so without adequate analysis. However, researchers only can infer reasons of failure by observing the results in limited circumstances. Most of the failures that RS

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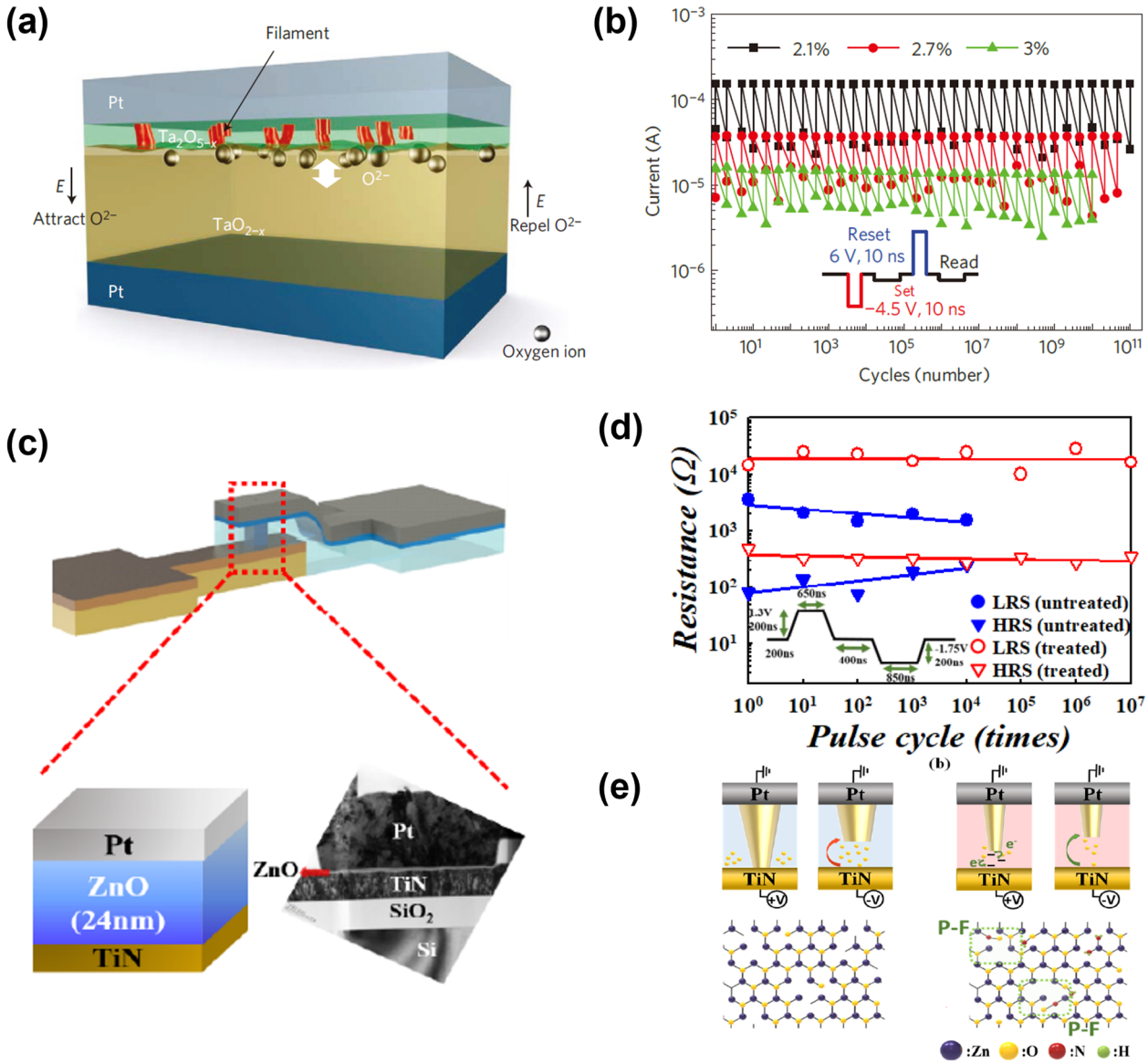


FIG. 3. (a) Schematic of the bilayer TaOx memory device. The movements of oxygen ions and vacancies are used to model resistive switching behavior. (b) Endurance characteristics of the TaOx bilayer deposited under 2.1%, 2.7%, 3.0% oxygen partial pressure at 0.5× 0.5 um^2 cell size; applied pulses are composed of −4.5 V, 10 ns for the set process and 6 V, 10 ns for the reset process. (c) 3D schematic structure of the Pt/ZnO/TiN RRAM device and cross-sectional TEM image of the device. (d) Endurance characteristics and an applied voltage waveform of measurement for the ammoniation annealed and intrinsic device. The untreated device fails resistive switching after E4 cycles. (e) Schematic diagrams of the ammonia doping process during the set and reset operation of (left) untreated and (right) ammoniation annealed devices. (a) and (b) Copyright 2011, Springer Nature. (c)–(e) Copyright 2020, IEEE.

devices show mainly originated from the changes in microstructure, such as the penetration of electrode metal atoms or vacancy forming in the RS layer, which permanently ruins the device operation into the LRS.[54,55] The thermal effect originated from the high current value of the LRS might be contributed to the failure of CFs, as they promote diffusion of the CF elements (vacancies, atoms, etc.). Moreover, sometimes the resistance of the device is unexpectedly trapped in a particular state and then returned to normal functioning again, which would be the case when too many ions or atoms move in one state transition and voltage is not enough to return them. If one reset operation displaces an excessive number of ions, voltage value of following set process might be insufficient, leading to stuck in HRS for some cycles. In addition, sometimes distribution of R_{HRS} and R_{LRS} suddenly changes, and in this case, the switching window between LRS and HRS may be much lower. Finally, stick in irreversible HRS might be the result of CF

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melting due to the high currents generated by LRS, which induces electromigration.

4. Endurance improvement through electrical measurement setting optimizations

Endurance enhancement can be achieved by the optimization of electrical measurement settings. Kim *et al.* fabricated a crossbar-type device with a TiO_2 resistive switching layer, whose CF consists of Magneli phases[56] and is formed and ruptured by an unidirectional motion of oxygen ions.[57] Researchers applied different voltage bias schemes composed of "PP;" each of the SET and RESET processes is operated by positive bias, "NN;" the same process with PP except a negative bias was applied and a modified bias scheme with a mixture of PP and NN (n × PP + n × NN) switching. The PP bias scheme enabled switching only up to ~140 cycles, while a mixture of the PP and NN bias schemes with n = 1, 3 showed more than 1500 successful RS. However, a series of PP and NN with n = 10 reported only 150 cycles of RS again. This phenomenon is due to minimized oxygen loss by drift and diffusion to Pt electrodes. After the PP–N operation, hourglass-shaped CF formed in a TiO_2 film, and subsequent negative bias would break the middle region of the CF, which leads to a decrease in oxygen loss compared to the PP–PP bias scheme. This is also consistent with 80%–90% of endurance degradation in the n = 10 PP and NN mixture bias scheme; larger n induces conical-shaped CF rupture, not hourglass shape, which leads to oxygen consumption over a large area.

In addition, Zhao *et al.* improved the number of switching cycles by programming pulse modulation on a TiN/ETML TaO_x/HfO_x/TiN stacked RRAM device.[58] Through switching window tuning, the worst endurance characteristic (10^5 cycles) that exhibited in full window (HRS 1 MΩ and LRS 20 kΩ) condition has been improved. Modified endurance characteristics have been reported in the moderate (HRS 1 MΩ, LRS 100 kΩ, 10^6 cycles) and narrow window (HRS 100 kΩ, LRS 20 kΩ, $>10^8$ cycles). Moreover, when the HRS was reduced from 1 MΩ to 100 kΩ and the LRS was fixed to 20 kΩ, the endurance was enhanced from 10^5 to 10^9. However, the current measurement revealed negligible change (less than one order of magnitude) when the LRS was changed from 20 to 100 kΩ and the HRS was fixed around 1 MΩ to 200 kΩ. Based on these measurements, the HRS was assumed to affect the endurance characteristics more significantly than the LRS. Therefore, RRAM devices should be controlled at a relatively low resistance range for better endurance characteristics.

5. Endurance improvement through RS layer doping

In addition to altering the resistance range, other methods have been adopted to enhance the endurance characteristics. Wu *et al.* constructed a Pt/ZnO/TiN RRAM device and doped the ZnO RS layer with ammonium to repair O and Zn vacancies using a low-temperature-high-pressure annealing process at 120 °C [Fig. 3(c)].[28] The treated device exhibited a wide memory window (untreated 0.9 order and treated 1.3 order), small operation voltage, and improved endurance characteristics with 10^7 stable RS cycles [Fig. 3(d)]. In the untreated device, large forming voltages induce a thick filament, leading to current increments in both the HRS and LRS. On the contrary, ammonium doping mitigates the current augmentation by repairing O and Zn vacancies and forms ZnO : NH_x crystals in the annealed device. The doped RS layer donates its electrons to fill the Zn vacancy, forming strong bonds and stabilizing the oxygen. Therefore, a low operating voltage is required in the doped device, leading to thin filament formation and endurance improvement [Fig. 3(e)].

Syu *et al.* constructed Pt/$WSiO_x$/TiN memory cells and investigated RS behavior when nitrogen doping is applied in $WSiO_x$ through co-sputtering SiO_2 and WSi_x in Ar/NH_3 mixed gas.[59] In consequence, $WSiO_x$/WSiON double switching layer devices showed stable RS for $>10^8$ times, while those of single-switching layer devices showed failure around ~10^6 cycles of switching. This enhancement through nitrogen doping is estimated to be originated from oxygen migration confinement of the introduced materials. In the doped device, the WSiON layer traps O_{2-} and localizes the oxygen around the CF due to the faster bonding speed of O_{2-} with nitride compared to oxide, as well as bonding energy between nitrogen and oxygen is higher than the oxygen–oxygen bond.

6. Endurance improvement through interlayer insertion

Interlayer insertion has also been introduced by many researchers as it can manipulate the movement of the CF forming material in the RS layer. Ismail *et al.* showed improvements in RS characteristics of CeO_2-based devices through thin Al interlayer insertion.[60] The interlayer was oxidized to Al_2O_3 extracting oxygen ions from the top and bottom electrodes, and it leads to the increment of more oxygen vacancies to CeO_2 layers by reducing the formation energy. In addition, endurance characteristics of Al doped devices showed stable for $>10^4$ dc switching cycles with a resistance ratio of ~10^2, while undoped device showed incomplete set/reset operations around only 200 cycles. This phenomenon is due to the filament thickness increase with RS cycles as the top electrode absorbs oxygen ions from the RS layer and forms the TiO interlayer, thereby creating oxygen vacancies, which expands CF. Therefore, a larger value of current compliance is necessary to rupture thicker CF and fill the vacancies for the stable SET/RESET process. However, Al-doped devices exhibited successful set and reset operations for 10^4 cycles because of oxygen vacancy CF randomness suppression[61] and oxygen vacancy migration hindering the role of Al_2O_3 interlayer.

Cao *et al.* proposed a back-end-of-line (BEOL) TiN barrier layer to stabilize the reset operation failure of the Cu/HfO_2/Ru memory device. Devices with the barrier layer showed 100% validity when 30 cycles of RS test applied to 20 devices, while those without the TiN layer showed only 20% validity. The negative-SET behavior in the Cu/HfO_2/Ru device originated from the Cu CF overgrowth into the Ru electrode. Therefore, an introduction of the TiN barrier layer suppressed the overgrowth of CF, which leads to the elimination of negative-SET behavior in the Cu/HfO_2/TiN/Ru memory device.

7. Endurance improvement through novel device structure

Constructing novel cell structures would be possible candidates for endurance enhancement methods. Tang *et al.* demonstrated a CMOS-compatible TiN/TaO_x/Hf/HfO_y/TiN RRAM stack, a connection of two back-to-back subcells.[62] Although each subcell can

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act as an RRAM device, the upper-cell only operates as a variable resistor. During the filament formation of the bottom-cell, the resistance of the upper TaO_x layer increased, stabilizing the current and size of the bottom-cell filament. On the contrary, the resistance of the upper TaO_x layer decreased during the HfO_x layer filament rupture, improving the device efficiency by enhancing V_{Oxygen} drift and diffusion. A hump was observed during the positive voltage sweep, indicating upper-cell reset, in which increased $R_{upper\text{-}cell}$ compensates for decreased $R_{bottom\text{-}cell}$. On the contrary, the current kept increasing at the initial stage and decreased after a specific point in the negative voltage sweep. Because of the decreased resistance of the upper-cell, most of the voltage was applied to the bottom-cell, resetting the bottom-cell with the decreased current. The back-to-back subcell structured RRAM device endured 2.1×10^4 DC cycles.

8. Insights into endurance

Endurance is the ability of the device to endure RS. It also refers to the number of cycles the device can sustain during the switching process. The methods of measuring endurance are diverse. However, if applying a write voltage takes too long, the endurance can be affected by the device retention characteristics.[50] Additionally, devices with poor endurance characteristics can suffer failure within a few cycles of the learning process.

C. Retention

1. Definition and measuring methods

Retention is the ability to endure voltage stress over time.[18,63] The retention of a RS device is evaluated by measuring the time the device sustains a constant current value without unacceptable decay, while a constant voltage stress is applied with a low read voltage (0.1 V).[64] Usually, a current–time plot is generated from the retention measurement. The standard value of retention required for a practical device is 10 years at 85 °C,[65] which is too long for practical measurement. Therefore, the retention value at 85 °C is extrapolated from the Arrhenius equation using the retention values at various temperatures,

$$Lifetime = B \times \exp\left(E_a/kT\right).$$

2. Retention in current devices

Many researchers have already reported RRAM devices exhibiting many years of retention ability at 85 °C (Fig. 4).[20,66] Yoon *et al.* fabricated an electrochemical metallization-like memristor composed of $Pt/Ta_2O_5/Ru$ layers [Fig. 4(a)] modulated by Ru atom-CF formation (ECM). Numerous experiments have verified that Ru is a filament-forming material. By comparing the current levels of the $Pt/Ta_2O_5/Ru$, $Pt/Ta_2O_5/Pt$, and $Pt/Ta_2O_5/Ta$ devices, the importance of the Ru bottom-electrode has been documented. Additionally, memristor devices with other switching materials (HfO_2, YSZ, and Al_2O_3) but the same electrodes exhibit similar RS behavior as a $Pt/Ta_2O_5/Ru$ device, suggesting that the Ru bottom-electrode is crucial in RS behavior. Finally, a sample with an opposite electrode structure ($Ru/Ta_2O_5/Pt$) exhibited the same electrical behavior as $Pt/Ta_2O_5/Ru$, except that the operation polarity is reversed, which indicates that Ru is the CF-forming material.

3. Retention failure analysis

Typically, retention failure occurs in the LRS as the HRS represents the original resistance value of the material. Therefore, manipulating the current compliance level has a profound impact on retention. Zhao *et al.* described a compact retention failure model of conductive-bridge random access memory based on physics (Fig. 5).[67] Retention failure originates from the lateral diffusion of metal atoms near the filament region. Additionally, the expansion of CF leads to the formation of an expansion region (ER); CF is divided into ER and residual effective CF (ECF). LRS degradation is interpreted as percolation path rupture due to the low density of the ER, while HRS degradation is construed as an unexpected percolation path formation in the ER (Fig. 5). Additionally, good agreement has been achieved between an actual $CuTeGe/Al_2O_3/TiN$ device's retention and the CF failure behavior of models at 100 °C.

4. Retention improvement through material optimization

Following the Arrhenius equation, the retention ability is greatly affected by the activation energy, a material-dependent value.

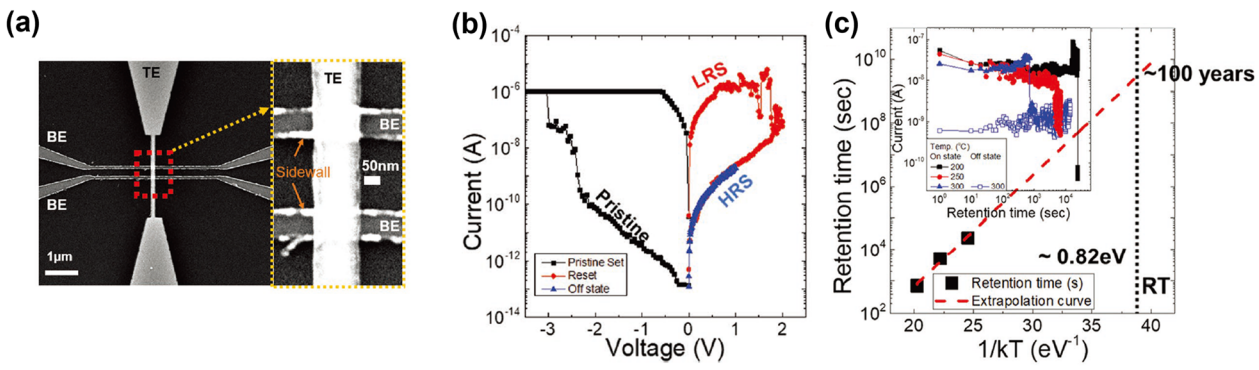


FIG. 4. (a) Top-view SEM image of the $Pt/Ta_2O_5/Ru$ nano-scale memristor. (b) I–V curves of the first set operation and following reset process. (c) Retention time vs 1/kT graph of Ru CF memristor (inset) retention time in three different temperatures from 200 to 300 °C. Copyright 2020, Wiley-VCH.

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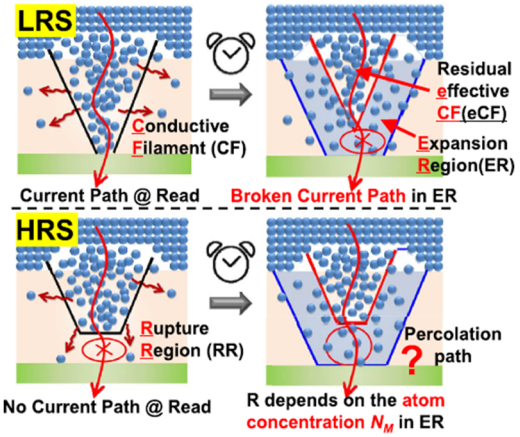


FIG. 5. Schematics of retention failure due to lateral diffusion. CF becomes wider over time and separates into two parts: the expansion region (ER) and the residual effective CF (eCF). Retention failure originates from percolation path rupture (LRS) and formation (HRS). Copyright 2019, IEEE.

Therefore, material selection is one of the fundamental solutions for improving retention. Moon *et al.* reported retention improvements in a Al/$Pr_{0.7}Ca_{0.3}MnO_3$ (PCMO)-based RRAM device by inserting a MoO_x buffer layer [Fig. 6(a)].[68,69] Due to the low activation energy (oxidation of the Al electrode) of the Al/PCMO device, the LRS retention was poor. However, inserting 3–4 nm of the MoO_x interlayer into the PCMO device enabled better retention times than initial devices by inserting other metal interlayers [Fig. 6(b)].

5. Retention improvement through nanostructure insertion

Many researchers have also achieved retention enhancement through nanocomposite embedding or process addition. Chen *et al.* embedded Ru nanocrystals (RuNCs) into Al_2O_3-based RRAM devices [Fig. 6(c)], and a constant voltage of 300 mV was applied to TaN/Al_2O_3/Pt and TaN/Al_2O_3 :RuNCs: Al_2O_3/Pt memory cells to estimate the retention ability of the devices.[30] The LRS and HRS of the devices with RuNCs were stable for 10^5 s, while poor HRS retention was observed in the pure Al_2O_3 sample [Fig. 6(d)]. This behavior originates from the local enhancement of the electric field, which induces filament rupture near the RuNCs.

6. Retention improvement through annealing

Adding certain processes during device fabrication can lead to retention enhancement.[37,38] Chen *et al.* applied an additional thermal budget during HfO_2/Hf 1T1R RRAM cell construction. Significant improvement in the retention from that of a non-annealed sample has been observed in both the LRS 10 and 40 μA current compliance levels. Through an additional annealing process, oxygen diffusion from the oxygen scavenging layer was limited, improving the device retention ability. RS behavior enhancement through Ar plasma irradiation interface engineering has been reported.

Ku *et al.*[38] deposited 9 nm-thick HfO_2 on a Pt (70 nm)/Ti (10 nm)/SiO_2 (300 nm)/Si substrate using ALD. The HfO_2 surface was irradiated by 50 W Ar plasma for 3 min at room temperature. The plasma-treated sample exhibited a retention time longer than that of the pristine device at 100 °C. The retention enhancement is due to the oxygen vacancy concentration difference; the high

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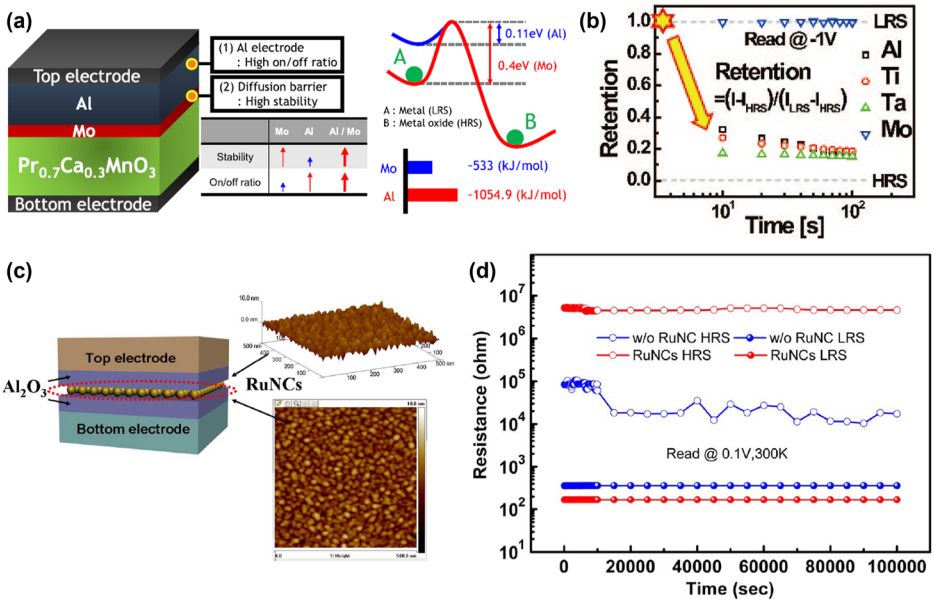


FIG. 6. (a) Schematics of the Al/Mo/PCMO device and its advantages (left) and the simple energy diagram of Mo insertion. (b) Normalized retention characteristics of the Al, Ti, Ta, and Mo inserted PCMO memristor at room temperature. (c) Schematic of the TaN/Al_2O_3 :RuNCs: Al_2O_3/Pt RRAM device and the AFM images of combined layer after deposition. (d) Retention characteristics of LRS and HRS at 300 K in the Al_2O_3 RRAM device with and without RuNC embedding.

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defect concentration at the interface of the Ar plasma-treated device induces retention improvement.

7. Insights into retention

Retention value is related to how long a device sustains a constant current value with an applied voltage pulse. Poor retention characteristics cause a loss of resistance state during weight modulation, which is detrimental to neuromorphic data processing. Retention is related to CF formation and rupture; thus, since the early 2010s, several material optimizations have been attempted to improve retention.[70–72] Typical examples include increasing the activation energy through new material layer insertion,[68,69] reducing the diffusion constant in the RS layer,[37] and increasing the electric field in the local area of the RS layer through nanostructure insertion.[30]

D. Synaptic linearity

1. Definition and effects on weight modulation

Synaptic linearity, regarded as one of the most important parts of neuromorphic computing, is the current response under voltage pulses.[72,73] Achieving linear conductance of memory device is necessary as it is directly related to weight modulation during ANN implementation.[74,75] Otherwise, the recognition accuracy significantly decreases, expending too much energy and time.[44,76]

2. Synaptic linearity improvement through pulse modulation

Pulse modulation is the most straightforward approach to achieve linear potentiation and depression characteristics.[24,36] Chen *et al.* proposed a novel programming scheme for linear weight updating of memristor device. Researchers constructed three different programming schemes and measured the experimental weights of each scheme in a TaO_x/TiO_2-based synaptic device [Figs. 7(a) and 7(b)]. Scheme A consists of simple pulse trains of potentiation and depression, scheme B is a pair of positive and negative voltage pulses from a split of a single pulse, and scheme C is a pulse duration varied set. As seen in Fig. 7(b), the synaptic linearity of each scheme is in the order of C > B > A due to the overshoot-cancel and weight update slow-down effect.

Wang *et al.* proposed a bipolar pulse-training scheme (BP scheme) by introducing an additional pulse with opposite polarity to the original one, preventing rapid conductance change. Their study has defined nonlinearity (NL) as

$$NL = \mathrm{Max}|G_{P(n)} - G_{D(n)}| \text{ for } n = 1 \text{ to } n,$$

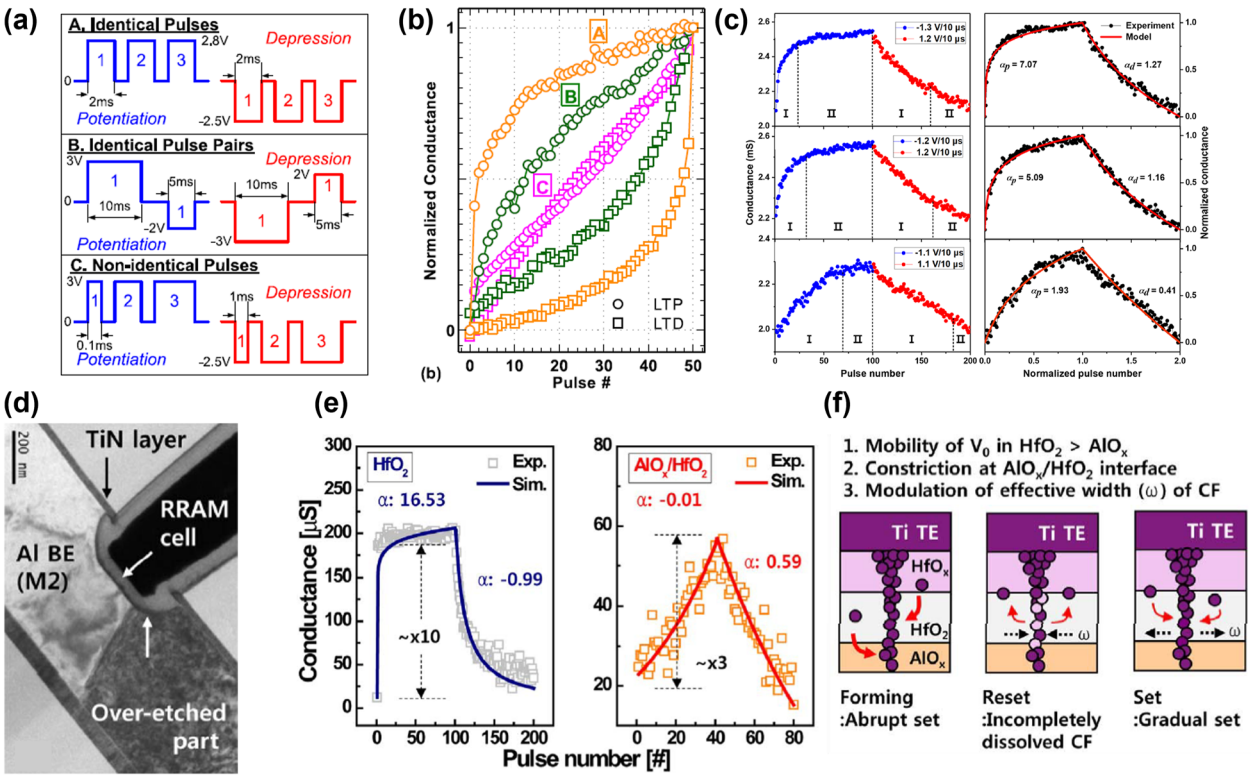


FIG. 7. (a) Three different programming schemes and (b) corresponding measured synaptic conductance data of the TaO_x/TiO_2 resistive synaptic device. (c) Conductance variations observed in the Cu^{2+} inserted KN memristor when 100 potentiation and depression pulses are applied. (d) TEM image of the 1T–1R array of the $Al/HfO_2/Ti/TiN$ stack memristor and (e) synaptic behaviors of $TiN/HfO_2/Ti/TiN$ stack (left) and $Al/HfO_2/Ti/TiN$ stack (right) devices. (f) Schematic illustration of a linear conductance modulation of $Al/HfO_2/Ti/TiN$ RRAM.

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where $G_{P(n)}$ and $G_{D(n)}$ are the conductance values after the n_{th} potentiation and depression pulses, respectively. In the measurements, when n = 50, the NL was 0.6–0.81 at n = 25 for the original pulses. In contrast, the proposed BP scheme led to NL = 0.42–0.54 under the same conditions. Therefore, it can be presumed that a weak pulse with opposite polarity mitigates the abrupt charge accumulation at the Ta/TaO_x interface. However, adopting a new programming scheme requires a read-before-write step to determine the state and adopt suitable pulse modulation to the device. This step increases the circuit design complexity, leading to excessive latency and energy consumption.

3. Synaptic linearity improvement through RS layer doping

RS layer doping can be a solution to the conductance nonideality of a memristor device.[27,77] Chandrasekaran *et al.* incorporated the Al dopant into a HfO_2-based memristor to improve the synaptic linearity.[77] Two kinds of devices were fabricated; one had an undoped HfO_x RS layer and the other had an Al-doped HfO_x switching layer comprising HfO_2 (2.5 nm)/Al_2O_3 (1 nm)/HfO_2 (2.5 nm). A double pulsing technique was adopted to measure the synaptic linearity. The first pulse had a width of 100 μs with a 2.5 V amplitude for potentiation (−2.4 V for depression); the second pulse had a width of 1 ms and a 1.5 V amplitude for potentiation (−1.5 V for depression), and the current was read simultaneously. The HfO_x device exhibited gradual current rise and fall after applying 90 repeated potentiation pulses and 80 depression pulses. However, the Al-doped HfO_2 device needed only 50 negative pulses to return to the value before the pulses were applied, with a higher conductance ratio than that of the undoped device. This synaptic linearity enhancement is due to the formation of separate O-poor and O-rich regions due to Al doping in the RS layer. Because the undoped device cannot form a strong filament due to scattered oxygen vacancies, the O-rich region of the doped HfO_x device ensures continuous and gradual growth and rupture of CF during potentiation and depression. RS layer doping is also one of the representative technologies used in perovskite-based memristors.

Park *et al.* suggested a simple method to improve conductance modulation.[27] Through Cu^{2+} doping in the $KNbO_3$ RS layer of a Pt/$KNbO_3$ (KN)/TiN/Si stacked memristor, the conductance linearity significantly improved due to Cu^{2+} replacing Nb^{5+} and producing oxygen vacancies. Cu^{2+} doping was accomplished by adding CuO to calcined KN powders under a pressure of 100 kgf/cm^2 and sintering at 940–980 °C for 1–2 h. The fabricated device exhibited various stages in the potentiation and depression curves [Fig. 7(c)]. Stage 1 represents the filament growth step, which is attributed to the redox process, while stage 2 is the region where oxygen vacancy diffusion controls the CF size. An intrinsic KN device exhibited a relatively small stage 1 region that widens with Cu^{2+} addition. This trend suggests that filament variation is mainly influenced by the redox process due to oxygen vacancies with high Cu^{2+} concentration. Current modulation linearity was enhanced, as observed in stage 1 region expansion, with increasing dopant materials. Considering the conductance shown in Fig. 7(c), the potentiation and depression curves occur at the LRS state, which means that RS behavior can be explained through the change in the memristor CF filament width.

Yeon *et al.* alloyed the conduction channels of a Si memristor device with Ag and Cu, which act as mobile metal atoms and RS stabilizers, respectively.[8] The RS characteristics were not observed in the pure Si device. However, the property was observed in the Ag-alloyed device as Ag is thermodynamically unstable and electrochemically mobile in Si. Additionally, 50 repetitions of 50 ns positive (negative) pulses were applied, and the asymmetric nonlinearity factor (ANL) was calculated to evaluate the potentiation (depression) characteristics of the Ag–Cu alloy and pure Ag devices. The average ANL of the Ag–Cu alloy devices was 0.30, while that of the pure Ag devices was 0.59. It was speculated that stochastic CF dissolution due to the thermodynamically unfavorable accommodation of Ag into Si led to a nonlinear weight update. On the contrary, alloying Si CF induced stabilized interaction, and the device exhibited uniform analog switching.

4. Synaptic linearity improvement through layer insertion

The previous approaches were also effective, but the most widely used method to improve synaptic linearity would be interlayer insertion.[36,78–82] Woo *et al.* demonstrated the response of identical pulses on two RRAM systems, TiN/HfO_2/Ti/TiN and Al/HfO_2/Ti/TiN stacks [Fig. 7(d)].[81] The only difference between the two devices was the bottom-electrode material, which induces Al_2O_3 thin-layer formation during HfO_2 ALD between the BE and RS layers. As shown in Fig. 7(e), the presence of thin interlayer greatly influenced the linearity of the conductance modulation. In order to determine the degree of improvement, the synaptic nonlinearity is expressed as α. A larger α value was obtained from the HfO_2-based memristor (16.53 for potentiation and −0.99 for depression) than from the device with AlO_x (−0.01 for potentiation and 0.59 for depression). These results can be explained by the mobility difference of the oxygen vacancies in the AlO_x and HfO_2 layers [Fig. 7(f)]. During the reset process, oxygen vacancy dissolution is avoided in the bilayer memristor with AlO_x due to preferation of CF width modulation through the HfO_2 layer. A lowered on/off ratio and continuously increasing current (instead of abrupt CF formation) have been observed due to the incomplete dissolution of CF in the bilayer system.

Wu *et al.*[80] focused on the thermal effect in the memristor to enhance device synaptic linearity. Because the fabricated memristor exhibited gradually increasing current at high temperatures, TaO_x was inserted into the HfO_x-based RRAM device as a thermal enhancement layer so that the device could exhibit analog switching behaviors. As expected, the HfO_x/TEL RRAM device exhibited excellent analog switching behavior at room temperature. Abrupt switching in the initial device can be explained by the positive feedback of the electric field. In contrast, the thermally enhanced RS layer accelerated multiple weak CF formations. Additionally, oxygen vacancies were distributed uniformly in the TEL-inserted RRAM device. Because the contribution of the formation and rupture of each filament from the total conductance change was negligible, the conductance did not change abruptly.

5. Insights into synaptic linearity

Synaptic linearity is the degree of linearity of conductance when constant voltage pulses are applied to the device. This property

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impacts learning simulations based on the ANN structure.[8,40] The simplest methods to improve synaptic linearity would be pulse modulation. However, this method may increase the circuit complexity as it needs a read-before-write process.[25,36] Therefore, numerous material modifications, including RS layer doping[27,77] and interlayer insertion,[36,78–82] have been proposed. These approaches are commonly employed to improve synaptic linearity; however, they sacrifice switching windows. Finally, a study that deploys the unique properties of the RS material confirmed linearity enhancement through selective etching[40] along the RS layer defects.

E. Multimodality

1. Definition and operation of a multimodal RRAM device

The conductive filament of a memristor originates from the movements of mobile ions, defects, and diffusive atoms in the switching layer, some of which are influenced by external stimulants, such as moisture, temperature, and light.[26,83,84] Through adopting materials with these properties by the RS layer, a memristor having multiple inputs; a multimodal memristor, can be fabricated. These devices can overcome the limitations in existing memristor devices, such as data acquisition isolation, complex hardware, and static coupling coefficients.[85] Among them, the most notable are memristors that respond to humidity or light.[33,41–43,58,84–86] If a photosensitive material is adopted as a switching layer of the memristor, the device would be a voltage–light bimodal memristor. Meanwhile, a voltage–humidity bimodal memristor can be fabricated by constructing an RS layer from an organic material that facilitates proton-coupled electron transfer (PCET).[87–89] In this Research Update, we clarified multimodal memristor if either the set or reset process of the device is mediated by a non-electrical input.

2. Light–voltage multimodal RRAM devices

As described above, researchers have employed photosensitive materials used in photovoltaics (PVs) as a memristor switching layer. Tan *et al.* constructed an ITO/CeO_{2-x}/AlO_y/Al device.[43] The device exhibits photosensitive characteristics owing to the presence of cerium oxide with a fluorite crystalline structure that provides extra defect energy levels in the bandgap and enables a wide wavelength window for photo response. The fabricated device exhibited RS characteristics (rectifying ratio of 10^4 at ±2 V) at 0 V → +2 V → 0 V → −2 V → 0 V voltage sweep and illumination-based LRS switching through 400–800 nm broadband emission on a halogen lamp at 60 pW μm^{-2} for 20 s with an on/off ratio of 30. Recently, research on hybrid organic–inorganic halide perovskite (HOIP) materials has been extended to optoelectronic memristor devices.

A representative of HOIP memristors is the $CsPbBr_3$ quantum dot (QD)-based photonic RRAM device reported by Wang *et al.*[90] The electrical characteristics, such as the turn-on voltage reduction and on-current enhancement (turn-on voltage decreasing from 2.6, 2.3, 1.7, and 1.4 to 1.1 V and on-current changing from 2.32×10^{-5}, 5.68×10^{-5}, 4.92×10^{-5}, and 5.29×10^{-5} to 4.58×10^{-5} A at 0, 0.041, 0.069, 0.129, and 0.153 mW cm^{-2} illumination, respectively), of the fabricated device changed due to an optical signal, suggesting that both light and the electric field can change the resistive states of the device significantly. The electrical characteristics of the device can be explained by the PV effect in $CsPbBr_3$ QDs. Electron–hole pairs generated from the QDs are separated and further arrested by the QDs to form an additional electrical field, inducing Br^- and Br vacancy formation, which forms the device CF.

3. Humidity–voltage multimodal RRAM devices

Humidity can also facilitate RS behavior. Song *et al.* fabricated a memristor using short-length peptide materials (YYACAYY, Y7C)[26,91] that induce PCET owing to their tyrosine-rich structure [Figs. 8(a) and 8(b)]. Due to PCET, the protons in the film facilitate redox reactions and metal atom accumulation, significantly influencing memristor CF formation. The current–voltage plot of the Y7C memristor varied with relative humidity (RH). When the RH was increased from 15% to 90%, the set voltage of the Y7C memristor decreased from 4.6 to 0.4 V, and a high HRS initial current was observed due to high proton conduction [Fig. 8(b)]. Moreover, unlike other bimodal memristors, both the set and reset processes were enabled in the Y7C memristor through a non-electrical input (RH). The authors of this research conducted three verifications to prove that the RS characteristics of Y7C memristors were mediated by proton conduction. The first was electrochemical impedance spectroscopy (EIS) analysis of the Y7C film to investigate the ions in the film; the second was kinetic isotope effect measurements[92] to analyze the changes in the electrical properties of D_2O vapor injection (instead of H_2O) in the chamber; and the last was comparing the switching characteristics of the F7C-based memristor, in which tyrosine (Y) in the RS layer was replaced with phenylalanine (F) in the original device. The EIS results, represented by Nyquist plots, showed a semicircle and a non-vertical line, indicating electrical double-layer formation that is attributed to ionic movement.[93–95] RS occurred during D_2O vapor injection, and a similar tendency was observed due to the RH level with that in H_2O, except higher set voltage, indicating a higher kinetic barrier for the Ag redox reaction and ionic conduction in the film. Therefore, the experiments proved that the hydrogen atoms from moisture considerably affect the memristor RS characteristics. Finally, the set voltages for the F7C and Y7C memristors were 10.8 and 1.8 V, respectively, suggesting that the tyrosine hydroxyl group reduces the energy barriers for Ag ions and promotes CF formation. Through the three verifications and RS measurements, it has been concluded that Y7C memristors exhibit proton-mediated RS that enables set and reset processes from both voltage and humidity.

4. Humidity, light, and voltage tri-modal RRAM devices

Wang *et al.* proposed that memristors can respond to three inputs[19] using the MXene nanosheet due to its proton/photon-mediated plasticity.[96] They demonstrated tri-modal in-sensor computing based on flexible memristor units using an MXene nanosheet/ZnO nanoparticle heterostructure [Fig. 8(c)]. The high hydrophilicity of the –OH bond-terminated Ti_3C_2 in the MXene integrates with the photo-active ZnO NPs, enabling multimodal sensing capability. Therefore, an RS behavior modulated by three

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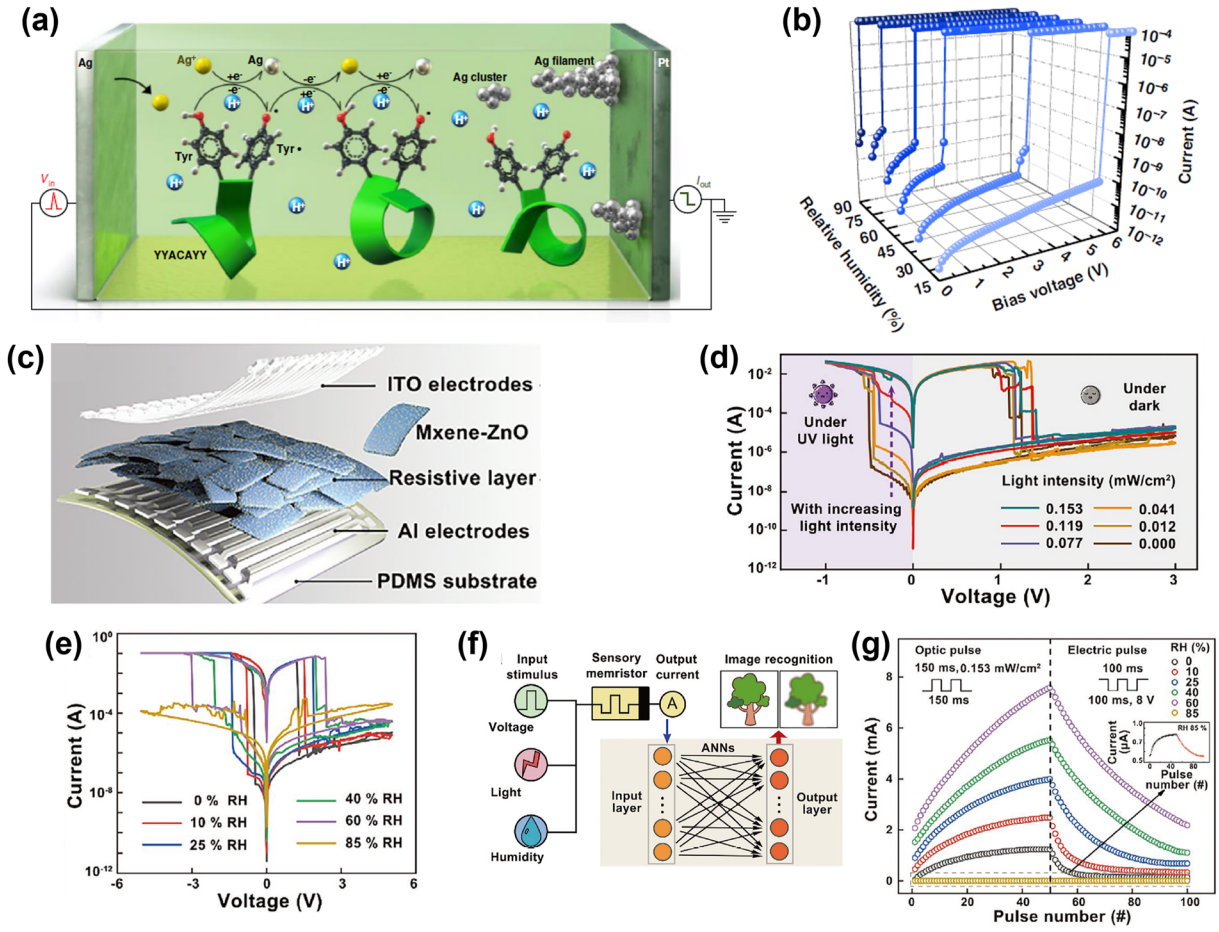


FIG. 8. (a) Schematic diagram RS of the Y7C film. Peptide backbones are expressed in green ribbons. (b) RS characteristics of the Y7C peptide under various RH conditions. (c) Schematic of the Mxene-based flexible RRAM device. I–V curves of the Mxene memristor device under (d) various light irradiance and (e) and RH conditions. (f) Schematic illustration of the multi-modal memristor based neuromorphic visual recognition system. (g) RH-mediated synaptic linearity of the Mxene-based memristor device under 50 optical pulses (long-term potentiation) and 50 electrical pulses (long-term depression).

inputs (voltage, humidity, and light) and proton/photon-mediated plasticity was observed during the measurement. Under ultraviolet (UV) illumination at 365 nm, the set voltage of the device decreased with light intensity. An LRS was achieved under light irradiation at 0.153 mW cm^{-2} even without any voltage input [Fig. 8(d)]. This can also be explained by photovoltaic operation. Absorbed UV photons with energies higher than the ZnO bandgap generate excitons; each charge carrier is separated in the heterostructured device. The electrons from here are trapped by the MXene, forming an internal electric field that induces LRS switching. The protonic device operation is represented in Fig. 8(e). Contrary to the Y7C memristor, the on/off ratio decays considerably under humid conditions due to proton obstruction by V_O CF. Finally, the researchers measured the potentiation and depression curves under varying RH conditions using 50 optical pulses (153 mW cm^{-2}, on and off times for 150 ms) for the set operation and 50 electrical pulses (−8 V, on and off time of 100 ms) for the reset process [Fig. 9(g)]. The lowest extracted nonlinearity value at 60% RH enables linear synaptic update and high recognition accuracy.

5. Insights into multimodality

Multimodal memristors refer to devices with tunable RS characteristics using non-electrical external inputs, such as light[41–43,58,84–86,90] and humidity.[26,91,97] These inputs significantly change the current value of the memristor, with the device exhibiting RS only through non-electrical inputs. After numerous memristors with various RS layers were reported, memristors with materials that exhibit the PV effect emerged in the middle of the 2010s.[98,99] Humidity–voltage bimodal memristors using RS layers in which PCET occurs were also presented. Multimodal modulation of memristor is significant in terms of emulating the human brain, which transmits the electrical signals by transporting numerous neurotransmitters and sometimes improving linearity.[87]

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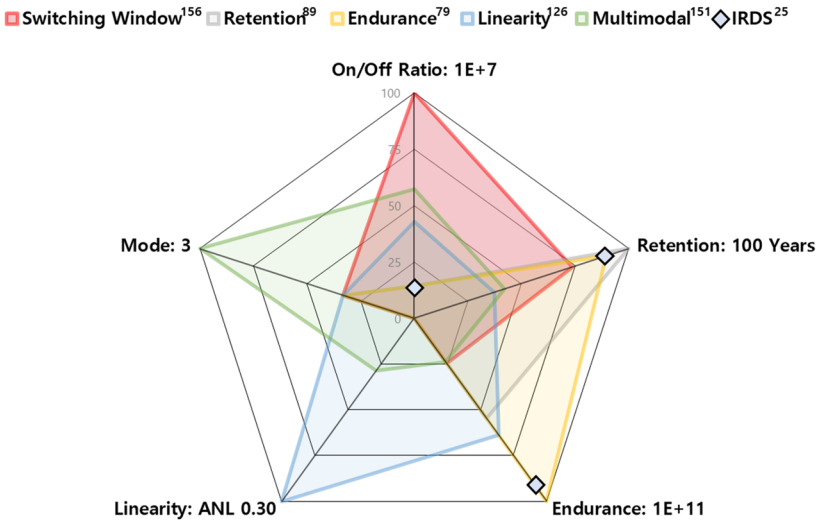


FIG. 9. Five typical parameters of RRAM devices showing exclusive performances in each aspect. Most devices show insufficient values in each of the multiple aspects.

III. CONCLUSIONS AND OUTLOOK

With the increasing applications of AI, numerous studies have reported RS devices that can implement the weight modulation of ANNs recently. Several devices have exhibited excellent performance for specific features; however, their progress is insufficient in the rest of the aspects, as shown in Fig. 9. As described above, deficits in certain features can cause failure in device operation; therefore, developing a versatility of memristor device is essential to commercialize those. In this Research Update, five factors that significantly influence the weight modulation of RRAM are discussed, and those values of RRAM are summarized in Table I. The factors that result in the best performance in each aspect are shown in Fig. 9 and are as follows:

1. Han *et al.* reported a switching window of 10^7 by introducing black phosphorous quantum dots between two poly (methyl methacrylate) polymer layers based on RRAM32. Despite the extraordinary switching ratio of the device, a retention time of 10 000 s and only 500 times RS availability were observed, which are below the requirements. Therefore, improvements are needed to meet the standard values. 2. A TaO_x bilayer-based RRAM device reported by Lee *et al.*[22] exhibited exceptional endurance characteristics (10^{12} times) and retention times (10^{11} s). However, a limited switching window leads to a small number of states in synaptic operation. This study was reported in 2011. Thus, it is presumed to be focused only on memory operation. Researchers may needed to concentrate only on retention and endurance. 3. Yoon *et al.* fabricated $Pt/Ta_2O_5/Ru$-structured memristive devices whose CFs were composed of Ru atoms.[21] The device exhibited good retention (almost 100 years at RT), low switching current (<10 μ A), and adequate retention time (10^6 s). However, its narrow window could be an obstacle to synaptic operation. 4. To enhance the synaptic linearity, Yeon *et al.* alloyed Ag and Cu into a Si-switching-medium memristor.[8] The Ag/Cu-alloyed memristor had an average ANL of 0.30, while the Ag-alloyed device showed an ANL of 0.59. Although a trade-off exists between the synaptic linearity and switching window, the Ag/Cu-alloyed device exhibited an HRS/LRS ratio of 10^3. However, the device was estimated to withstand only for 1 h at 85 °C, which could act as a glitch during weight modulation. 5. Finally, a tri-modal MXene–ZnO memristor was reported by Wang *et al.*[19] Memristors operating with more than two inputs are impressive. However, the device is not superior to other requirements except for multimodality. The devices exhibit limited lifetimes and cycles, insufficient switching ratios, and nonlinear conductance modulation. We anticipate improvements within a few years as only some reports have so far succeeded in fabricating tri-modal memristors.

In recent years, studies on RS-based RRAM have achieved remarkable advances due to the numerous efforts of material science scholars. Therefore, the alteration of conventional memory devices with low speed and large computing power to neuromorphic devices is now enabled, which allows for an extension in commercial applications of the neuromorphic device, such as image recognition, autonomous vehicles, and data analysis. Yet, still limits exist. There are several requirements[18] for the commercialization of RRAM, and some prior reports have already succeeded in achieving the single criteria. However, there are only a few reports that have shown acceptable performances for multiple requirements simultaneously. Therefore, a prospective and future development of the RRAM device would be achieving multi-requirements together without missing any aspects. Currently, the reports covered in this Research Update tend to be biased to only one aspect, as shown in Fig. 9. Although satisfying every aspect (switching window, endurance, retention, synaptic linearity, and muilti-modality) would be difficult due to the trade-off relationship between some features, it is important to develop devices that sufficiently satisfy various features unless it would degrade performance of the neural network.

By summarizing the efforts to improve the various requirements of RRAM devices, we believe that this Research Update can

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TABLE I. Summary of the RRAM requirement value (switching window, endurance, retention, synaptic linearity, and multi-modality).

Structure	Switching ratio	Retention (s)	Endurance	Linearity	Multi-modality	Reference
Au/ZnO nanorod/AZO	10^4	···	100 cycle	···	Mono	39
TiN/HfO_x/Al_2O_3/graphene	~10^2	10^5 (423 K)	1600 cycle	···	Mono	100
ITO/HfO_x/MoS_2–Pd NPs/ITO	1600	10^4	200 cycle	···	Mono	33
Ag/dislocation-free i-Si/p-Si	10^4	1.87 yrs at RT	10^9	···	Mono	40
Pt/ZnO/IZO	912	10^5	10^5	···	Mono	34
Pt/ZnO/TiN	1.3 order	10^4	10^7	···	Mono	28
$Hf_{0.1}In_{0.42}Sn_{0.06}O_{0.42}$/$ZrO_2$/TiN	300	10^7	10^4	···	Mono	47
Ti/HfO_x/NI/HfO_x/Pt	590	···	···	···	Mono	35
PET/Al/PMMA/BPQDs/PMMA/Al	10^7	10^4	500	···	Mono	21
Ag/PPX-Ag/ITO	10^3	10^4	10^4	···	Mono	101
FTO/$CH_3NH_3PbCl_xI_{3-x}$/Au or Ag	10^9	···	···	···	Mono	102
Pt/TaO_x/Pt	10	10 yrs at 85 °C	>10^9	···	Mono	66
Pt/Ta_2O_5/Ru	10	100 yrs at 85 °C	10^6	···	Mono	20
HfO_x/Hf 1T1R	10^2	3.6×10^6	···	···	Mono	37
TaN/Al_2O_3:RuNCs:Al_2O_3/Pt	10^5	10^5	···	···	Mono	30
Ti/HfO_2/Pt	19.2	2×10^4 at 100 °C	6.6×10^4	···	Mono	38
Pt/TaO_x/HfO_2/Pt	1350	3.6 yrs at 85 °C	150	···	Mono	103
Mo/$Pr_{0.7}Ca_{0.3}MnO_3$	>10^2	>10^4	···	···	Mono	68
Cu/nanohole-graphene/HfO_2/Pt	10^5	2×10^5 at 125 °C	10^7	···	Mono	104
Pt/Ta_2O_{5-x}/$Ta_{2-x}O_5$/Pt	10	10 yrs at 85 °C	10^{11}	···	Mono	22
Pt/Ta/TaO_x/Pt/Ta	5	···	10^6	···	Mono	105
Al/ZnO/Al	10	10^6	250	···	Mono	106
TiN/ETML/HfO_x/TiN	12	···	10^{11}	···	Mono	58
TiN/TaO_x/Hf/HfO_y/TiN	10	···	2×10^4	···	Mono	62
Pt/TiO_2/Pt	10^2	>10^3	···	···	Mono	57
TiN/Ti/HfO_x/TiN	10	···	10^8	···	Mono	107
IrO_x/Al_2O_3/IrO_x-ND/Al_2O_3/WO_x/W	10^5	10 yrs at 85 °C	10^5	···	Mono	108
Pt/$WSiO_x$/WSiON/TiN	>10^2	10^5	10^8	···	Mono	59
Ti/CeO_2:Al/Pt	~10^2	10^5	10^4	···	Mono	60
Cu/HfO_2/TiN/Ru	~10^1	10^7	10^8	···	Mono	109
TaN/CeO_2/Ti/CeO_2/Pt	>10^3	10^4	10^4	···	Mono	110
Cu/SiO_2/Pt	>10^4	···	10^4	···	Mono	111
Ta/TaO_x/TiO_x/Ti	···	···	···	A = −3	Mono	25
Ta/TaO_x/TiO_x/Ti	···	···	···	NL = 0.42 ~0.54	Mono	36
TiN/TaO_x/HfO_x/TiN	···	···	···	A = 0.63	Mono	112
TiN/Ti/Al:HfO_2/TiN/TaN	10	10^4 in 180 °C	···	P 22%, D 60%	Mono	77
Pt/$KNbO_3$/TiN/Si	···	···	···	α = 1.93(P), 0.41(D)	Mono	29
Au/Cr/a-Si/Au/a-Si/p + Si	>10^3	1 h in 85 °C	10^7	NL = 0.30	Mono	8
TiN/SiO_2/TaO_x/Pt	3	···	10^6	β = 0.3(P), 0.2(D)	Mono	113
TiN/HfO_2/Ti/TiN	3	···	···	α = −0.01(P), 0.59(D)	Mono	81
TiN/HfO_x/Ti/Ta	···	···	···	98.4% (P), 98.1% (D)	Mono	78
ITO/CeO_{2-x}/AlO_y/Al	10^4	10^4	30	···	Bi	43
Au/$MaPbBr_3$/ITO	10^3	10^4	10^3	···	Bi	86
Au or Ag/$MAPbI_3$/Au	10^6	44 in RT	10^3	···	Bi	41
Ag/PMMA/$CsPbBr_3$/PMMA/ITO	10^5	4×10^5	5000	···	Bi	90
Al/GO-TiO_2/ITO	10^2	10^5 at 80 °C	500	···	Bi	114
Ag/Y7C/Pt	10^6	10^4	10^2	···	Bi	26
ITO/MXene-ZnO/Al	10^4	10^4	400	NL = 1.050	Tri	19

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guide researchers in fabricating RRAM devices for neuromorphic electronics to construct devices with satisfying values of numerous requirements because it is desirable for implementing superior neuromorphic computing chips.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jeong Hyun Yoon: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Young-Woong Song**: Formal analysis (supporting). **Wooho Ham**: Investigation (supporting). **Jeong-Min Park**: Visualization (supporting). **Jang-Yeon Kwon**: Conceptualization (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

REFERENCES

[1] A. Gandomi and M. Haider, "Beyond the hype: Big data concepts, methods, and analytics," Int. J. Inf. Manag. **35**, 137–144 (2015).
[2] T. N. Theis and H.-S. P. Wong, "The end of Moore's law: A new beginning for information technology," Comput. Sci. Eng. **19**, 41–50 (2017).
[3] M. Horowitz, "1.1 computing's energy problem (and what we can do about it)," in *2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC)* (IEEE, 2014), pp. 10–14.
[4] X. Liu, F. Wang, J. Su, Y. Zhou, and S. Ramakrishna, "Bio-inspired 3D artificial neuromorphic circuits," Adv. Funct. Mater. **32**, 2113050 (2022).
[5] W. Zhang, R. Mazzarello, M. Wuttig, and E. Ma, "Designing crystallization in phase-change materials for universal memory and neuro-inspired computing," Nat. Rev. Mater. **4**, 150–168 (2019).
[6] Y. LeCun, Y. Bengio, and G. Hinton, Nature **521**, 436–444 (2015).
[7] N. Yang, J. Zhang, J.-K. Huang, Y. Liu, J. Shi, Q. Si, J. Yang, and S. Li, "Multitasking memristor for high performance and ultralow power artificial synaptic device application," ACS Appl. Electron. Mater. **4**, 3154–3165 (2022).
[8] H. Yeon, P. Lin, C. Choi, S. H. Tan, Y. Park, D. Lee, J. Lee, F. Xu, B. Gao, H. Wu *et al.*, "Alloying conducting channels for reliable neuromorphic computing," Nat. Nanotechnol. **15**, 574–579 (2020).
[9] M. Onen, T. Gokmen, T. K. Todorov, T. Nowicki, J. A. Del Alamo, J. Rozen, W. Haensch, and S. Kim, "Neural network training with asymmetric crosspoint elements," Front. Artif. Intell. **5**, 891624 (2022).
[10] C. Lee, K. Noh, W. Ji, T. Gokmen, and S. Kim, "Impact of asymmetric weight update on neural network training with tiki-taka algorithm," Front. Neurosci. **15**, 1554 (2021).
[11] C. Choi, H. Kim, J.-H. Kang, M.-K. Song, H. Yeon, C. S. Chang, J. M. Suh, J. Shin, K. Lu, B.-I. Park *et al.*, "Reconfigurable heterogeneous integration using stackable chips with embedded artificial intelligence," Nat. Electron. **5**, 386–393 (2022).
[12] S.-O. Park, H. Jeong, J. Park, J. Bae, and S. Choi, "Experimental demonstration of highly reliable dynamic memristor for artificial neuron and neuromorphic computing," Nat. Commun. **13**, 2888 (2022).
[13] D. H. Kim, Y. S. Kim, W. H. Cheong, H. Song, H. Rhee, S. N. Kay, J.-W. Han, and K. M. Kim, "Evolutionary learning of binary neural network using a TaO_x memristor via stochastic stateful logic," Adv. Intell. Syst. **4**, 2200058 (2022).
[14] T. Ohno, T. Hasegawa, T. Tsuruoka, K. Terabe, J. K. Gimzewski, and M. Aono, "Short-term plasticity and long-term potentiation mimicked in single inorganic synapses," Nat. Mater. **10**, 591–595 (2011).
[15] X. Xu, Q. Luo, T. Gong, H. Lv, S. Long, Q. Liu, S. S. Chung, J. Li, and M. Liu, "Fully CMOS compatible 3D vertical RRAM with self-aligned self-selective cell enabling sub-5 nm scaling," in *2016 IEEE Symposium on VLSI Technology* (IEEE, 2016), pp. 1–2.
[16] X. Cui, X. Li, M. Zhang, and X. Cui, "Design of the RRAM-based polymorphic look-up table scheme," IEEE J. Electron Devices Soc. **7**, 949–953 (2019).
[17] C. Sun, S. M. Lu, F. Jin, W. Q. Mo, J. L. Song, and K. F. Dong, "Multi-factors induced evolution of resistive switching properties for $TiN/Gd_2O_3/Au$ RRAM devices," J. Alloys Compd. **816**, 152564 (2020).
[18] M. Moore, International roadmap for devices and systems (accessed January 2020).
[19] Y. Wang, Y. Gong, L. Yang, Z. Xiong, Z. Lv, X. Xing, Y. Zhou, B. Zhang, C. Su, Q. Liao, and S. Han, "MXene-ZnO memristor for multimodal in-sensor computing," Adv. Funct. Mater. **31**, 2100144 (2021).
[20] J. H. Yoon, J. Zhang, P. Lin, N. Upadhyay, P. Yan, Y. Liu, Q. Xia, and J. J. Yang, "A low-current and analog memristor with Ru as mobile species," Adv. Mater. **32**, 1904599 (2020).
[21] S. Han, L. Hu, X. Wang, Y. Zhou, Y. Zeng, S. Ruan, C. Pan, and Z. Peng, "Black phosphorus quantum dots with tunable memory properties and multilevel resistive switching characteristics," Adv. Sci. **4**, 1600435 (2017).
[22] M.-J. Lee, C. B. Lee, D. Lee, S. R. Lee, M. Chang, J. H. Hur, Y.-B. Kim, C.-J. Kim, D. H. Seo, S. Seo, U. I. Chung, I. K. Yoo, and K. Kim, "A fast, high-endurance and scalable non-volatile memory device made from asymmetric Ta_2O_{5-x}/TaO_{2-x} bilayer structures," Nat. Mater. **10**, 625–630 (2011).
[23] M. Lanza, A. Sebastian, W. D. Lu, M. Le Gallo, M. F. Chang, D. Akinwande, F. M. Puglisi, H. N. Alshareef, M. Liu, and J. B. Roldan, "Memristive technologies for data storage, computation, encryption, and radio-frequency communication," Science **376**, eabj9979 (2022).
[24] A. Chen, "A review of emerging non-volatile memory (NVM) technologies and applications," Solid-State Electron. **125**, 25–38 (2016).
[25] P.-Y. Chen, B. Lin, I.-T. Wang, T.-H. Hou, J. Ye, S. Vrudhula, J.-s. Seo, Y. Cao, and S. Yu, "Mitigating effects of non-ideal synaptic device characteristics for on-chip learning," in *2015 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)* (IEEE, 2015), pp. 194–199.
[26] M.-K. Song, S. D. Namgung, D. Choi, H. Kim, H. Seo, M. Ju, Y. H. Lee, T. Sung, Y.-S. Lee, K. T. Nam, and J. Y. Kwon, "Proton-enabled activation of peptide materials for biological bimodal memory," Nat. Commun. **11**, 5896 (2020).
[27] Y. Y. Chen, R. Roelofs, A. Redolfi, R. Degraeve, D. Crotti, A. Fantini, S. Clima, B. Govoreanu, M. Komura, L. Goux *et al.*, "Tailoring switching and endurance/retention reliability characteristics of HfO_2/Hf RRAM with Ti, Al, Si dopants," in *2014 Symposium on VLSI Technology (VLSI-Technology): Digest of Technical Papers* (IEEE, 2014), pp. 1–2.
[28] P.-Y. Wu, H.-X. Zheng, C.-C. Shih, T.-C. Chang, W.-J. Chen, C.-C. Yang, W.-C. Chen, M.-C. Tai, Y.-F. Tan, H.-C. Huang, X. H. Ma, Y. Hao, T. M. Tsai, and S. M. Sze, "Improvement of resistive switching characteristics in zinc oxide-based resistive random access memory by ammoniation annealing," IEEE Electron Device Lett. **41**, 357–360 (2020).
[29] S.-M. Park, H.-G. Hwang, J.-U. Woo, W.-H. Lee, S.-J. Chae, and S. Nahm, "Improvement of conductance modulation linearity in a Cu^{2+}-doped $KNbO_3$ memristor through the increase of the number of oxygen vacancies," ACS Appl. Mater. Interfaces **12**, 1069–1077 (2019).
[30] L. Chen, H.-Y. Gou, Q.-Q. Sun, P. Zhou, H.-L. Lu, P.-F. Wang, S.-J. Ding, and D. Zhang, "Enhancement of resistive switching characteristics in Al_2O_3-based RRAM with embedded ruthenium nanocrystals," IEEE Electron Device Lett. **32**, 794–796 (2011).

02 May 2026 09:56:35

© Author(s) 2023

[31] L. Zhao, S.-G. Park, B. Magyari-Köpe, and Y. Nishi, "Dopant selection rules for desired electronic structure and vacancy formation characteristics of TiO_2 resistive memory," Appl. Phys. Lett. **102**, 83506 (2013).
[32] N. Sedghi, H. Li, I. F. Brunell, K. Dawson, R. J. Potter, Y. Guo, J. T. Gibbon, V. R. Dhanak, W. D. Zhang, J. F. Zhang, J. Robertson, S. Hall, and P. R. Chalker, "The role of nitrogen doping in ALD Ta_2O_5 and its influence on multilevel cell switching in RRAM," Appl. Phys. Lett. **110**, 102902 (2017).
[33] X.-F. Wang, H. Tian, H.-M. Zhao, T.-Y. Zhang, W.-Q. Mao, Y.-C. Qiao, Y. Pang, Y.-X. Li, Y. Yang, and T.-L. Ren, "Interface engineering with MoS_2–Pd nanoparticles hybrid structure for a low voltage resistive switching memory," Small **14**, 1702525 (2018).
[34] S.-S. Li and Y.-K. Su, "Improvement of the performance in Cr-doped ZnO memory devices via control of oxygen defects," RSC Adv. **9**, 2941–2947 (2019).
[35] J. Wang, L. Li, H. Huyan, X. Pan, and S. S. Nonnenmann, "Highly uniform resistive switching in HfO_2 films embedded with ordered metal nanoisland arrays," Adv. Funct. Mater. **29**, 1808430 (2019).
[36] I.-T. Wang, C.-C. Chang, L.-W. Chiu, T. Chou, and T.-H. Hou, "3D Ta/TaO_x/TiO_2/Ti synaptic array and linearity tuning of weight update for hardware neural network applications," Nanotechnology **27**, 365204 (2016).
[37] Y. Y. Chen, M. Komura, R. Degraeve, B. Govoreanu, L. Goux, A. Fantini, N. Raghavan, S. Clima, L. Zhang, and A. Belmonte, "Improvement of data retention in HfO_2/Hf 1T1R RRAM cell under low operating current," in *2013 IEEE International Electron Devices Meeting* (IEEE, 2013), pp. 10–11.
[38] B. Ku, Y. Abbas, A. S. Sokolov, and C. Choi, "Interface engineering of ALD HfO_2-based RRAM with Ar plasma treatment for reliable and uniform switching behaviors," J. Alloys Compd. **735**, 1181–1188 (2018).
[39] Y. Sun, X. Yan, X. Zheng, Y. Liu, Y. Zhao, Y. Shen, Q. Liao, and Y. Zhang, "High on–off ratio improvement of ZnO-based forming-free memristor by surface hydrogen annealing," ACS Appl. Mater. Interfaces **7**, 7382–7388 (2015).
[40] S. Choi, S. H. Tan, Z. Li, Y. Kim, C. Choi, P.-Y. Chen, H. Yeon, S. Yu, and J. Kim, "SiGe epitaxial memory for neuromorphic computing with reproducible high performance based on engineered dislocations," Nat. Mater. **17**, 335–340 (2018).
[41] X. Zhu, J. Lee, and W. D. Lu, "Iodine vacancy redistribution in organic–inorganic halide perovskite films and resistive switching effects," Adv. Mater. **29**, 1700527 (2017).
[42] G. Chen, G. Peng, X. Yu, W. Yu, Y. Hao, Y. Dai, H. Chen, and T. Guo, "Multifunctional memory-synaptic hybrid optoelectronic transistors for neuromorphic computing," IEEE Trans. Electron Devices **69**, 3997 (2022).
[43] H. Tan, G. Liu, X. Zhu, H. Yang, B. Chen, X. Chen, J. Shang, W. D. Lu, Y. Wu, and R. Li, "An optoelectronic resistive switching memory with integrated demodulating and arithmetic functions," Adv. Mater. **27**, 2797–2803 (2015).
[44] M. Zhao, B. Gao, J. Tang, H. Qian, and H. Wu, "Reliability of analog resistive switching memory for neuromorphic computing," Appl. Phys. Rev. **7**, 11301 (2020).
[45] S. Yu, "Neuro-inspired computing with emerging nonvolatile memorys," Proc. IEEE **106**, 260–285 (2018).
[46] L. Li, K.-C. Chang, C. Ye, X. Lin, R. Zhang, Z. Xu, Y. Zhou, W. Xiong, and T.-P. Kuo, "An indirect way to achieve comprehensive performance improvement of resistive memory: When hafnium meets ITO in an electrode," Nanoscale **12**, 3267–3272 (2020).
[47] B. Govoreanu, A. Redolfi, L. Zhang, C. Adelmann, M. Popovici, S. Clima, H. Hody, V. Paraschiv, I. P. Radu, and A. Franquet, "Vacancy-modulated conductive oxide resistive RAM (VMCO-RRAM): An area-scalable switching current, self-compliant, highly nonlinear and wide on/off-window resistive switching cell," in *2013 IEEE International Electron Devices Meeting* (IEEE, 2013), pp. 10–12.
[48] P.-Y. Gu, Y.-S. Chen, H.-Y. Lee, P.-S. Chen, W.-H. Liu, W.-S. Chen, Y.-Y. Hsu, F. Chen, and M.-J. Tsai, "Scalability with silicon nitride encapsulation layer for Ti/HfO_x pillar RRAM," in *Proceedings of the 2010 International Symposium on VLSI Technology, System And Application* (IEEE, 2010), pp. 146–147.
[49] F. Zahoor, T. Z. Azni Zulkifli, and F. A. Khanday, "Resistive random access memory (RRAM): An overview of materials, switching mechanism, performance, multilevel cell (mlc) storage, modeling, and applications," Nanoscale Res. Lett. **15**, 90 (2020).
[50] M. Lanza, H. S. P. Wong, E. Pop, D. Ielmini, D. Strukov, B. C. Regan, L. Larcher, M. A. Villena, J. J. Yang, and L. Goux, "Recommended methods to study resistive switching devices," Adv. Electron. Mater. **5**, 1800143 (2019).
[51] N. Xiao, M. A. Villena, B. Yuan, S. Chen, B. Wang, M. Eliáš, Y. Shi, F. Hui, X. Jing, A. Scheuermann, K. Tang, P. C. McIntyre, and M. Lanza, "Resistive random access memory cells with a bilayer TiO_2/SiO_X insulating stack for simultaneous filamentary and distributed resistive switching," Adv. Funct. Mater. **27**, 1700384 (2017).
[52] J. J. Yang, M.-X. Zhang, J. P. Strachan, F. Miao, M. D. Pickett, R. D. Kelley, G. Medeiros-Ribeiro, and R. S. Williams, "High switching endurance in TaO_x memristive devices," Appl. Phys. Lett. **97**, 232102 (2010).
[53] M. Lanza, R. Waser, D. Ielmini, J. J. Yang, L. Goux, J. Suñe, A. J. Kenyon, A. Mehonic, S. Spiga, and V. Rana, "Standards for the characterization of endurance in resistive switching devices," ACS Nano **15**, 17214–17231 (2021).
[54] F. Palumbo, C. Wen, S. Lombardo, S. Pazos, F. Aguirre, M. Eizenberg, F. Hui, and M. Lanza, "A review on dielectric breakdown in thin dielectrics: Silicon dioxide, high-*k*, and layered dielectrics," Adv. Funct. Mater. **30**, 1900657 (2020).
[55] S. Balatti, S. Ambrogio, Z. Wang, S. Sills, A. Calderoni, N. Ramaswamy, and D. Ielmini, "Voltage-controlled cycling endurance of HfO_x-based resistive-switching memory," IEEE Trans. Electron Devices **62**, 3365–3372 (2015).
[56] D.-H. Kwon, K. M. Kim, J. H. Jang, J. M. Jeon, M. H. Lee, G. H. Kim, X.-S. Li, G.-S. Park, B. Lee, S. Han, M. Kim, and C. S. Hwang, "Atomic structure of conducting nanofilaments in TiO_2 resistive switching memory," Nat. Nanotechnol. **5**, 148–153 (2010).
[57] G. Hwan Kim, J. Ho Lee, J. Yeong Seok, S. Ji Song, J. Ho Yoon, K. Jean Yoon, M. Hwan Lee, K. Min Kim, H. Dong Lee, S. Wook Ryu, T. Joo Park, and C. Seong Hwang, "Improved endurance of resistive switching TiO_2 thin film by hourglass shaped Magnéli filaments," Appl. Phys. Lett. **98**, 262901 (2011).
[58] M. Zhao, H. Wu, B. Gao, X. Sun, Y. Liu, P. Yao, Y. Xi, X. Li, Q. Zhang, and K. Wang, "Characterizing endurance degradation of incremental switching in analog RRAM for neuromorphic systems," in *2018 IEEE International Electron Devices Meeting (IEDM)* (IEEE, 2018), pp. 20–22.
[59] Y.-E. Syu, R. Zhang, T.-C. Chang, T.-M. Tsai, K.-C. Chang, J.-C. Lou, T.-F. Young, J.-H. Chen, M.-C. Chen, Y.-L. Yang, C. C. Shih, T. J. Chu, J. Y. Chen, C. H. Pan, Y. T. Su, H. C. Huang, D. S. Gan, and S. M. Sze, "Endurance improvement technology with nitrogen implanted in the interface of $WSiO_x$ resistance switching device," IEEE Electron Device Lett. **34**, 864–866 (2013).
[60] M. Ismail, E. Ahmed, A. Rana, F. Hussain, I. Talib, M. Nadeem, D. Panda, and N. Shah, "Improved endurance and resistive switching stability in ceria thin films due to charge transfer ability of al dopant," ACS Appl. Mater. Interfaces **8**, 6127–6136 (2016).
[61] M. Soo Lee, S. Choi, C.-H. An, and H. Kim, "Resistive switching characteristics of solution-deposited gd, dy, and ce-doped ZrO_2 films," Appl. Phys. Lett. **100**, 143504 (2012).
[62] Y. Z. Tang, Z. Fang, X. P. Wang, B. B. Weng, Z. X. Chen, and G. Q. Lo, "A novel RRAM stack with TaO_x/HfO_y double-switching-layer configuration showing low operation current through complimentary switching of back-to-back connected subcells," IEEE Electron Device Lett. **35**, 627–629 (2014).
[63] F. Hui, E. Grustan-Gutierrez, S. Long, Q. Liu, A. K. Ott, A. C. Ferrari, and M. Lanza, "Graphene and related materials for resistive random access memories," Adv. Electron. Mater. **3**, 1600195 (2017).
[64] Y. Shi, C. Pan, V. Chen, N. Raghavan, K. L. Pey, F. M. Puglisi, E. Pop, H.-S. Wong, and M. Lanza, "Coexistence of volatile and non-volatile resistive switching in 2D h-BN based electronic synapses," in *2017 IEEE International Electron Devices Meeting (IEDM)* (IEEE, 2017), pp. 4–5.
[65] A. Pirovano, A. Redaelli, F. Pellizzer, F. Ottogalli, M. Tosi, D. Ielmini, A. L. Lacaita, and R. Bez, "Reliability study of phase-change nonvolatile memories," IEEE Trans. Device Mater. Reliab. **4**, 422–427 (2004).
[66] Z. Wei, Y. Kanzawa, K. Arita, Y. Katoh, K. Kawai, S. Muraoka, S. Mitani, S. Fujii, K. Katayama, and M. Iijima, "Highly reliable TaO_x ReRAM and direct evidence of redox reaction mechanism," in *2008 IEEE International Electron Devices Meeting* (IEEE, 2008), pp. 1–4.
[67] Y. Zhao, P. Huang, Z. Zhou, C. Liu, S. Qin, L. Liu, X. Liu, H.-S. P. Wong, and J. Kang, "A physics-based compact model for CBRAM retention behaviors based on atom transport dynamics and percolation theory," IEEE Electron Device Lett. **40**, 647–650 (2019).

02 May 2026 09:56:35

© Author(s) 2023

[68]K. Moon, E. Cha, J. Park, S. Gi, M. Chu, K. Baek, B. Lee, S. Oh, and H. Hwang, "High density neuromorphic system with Mo/$Pr_{0.7}Ca_{0.3}MnO_3$ synapse and NbO_2 IMT oscillator neuron," in *2015 IEEE International Electron Devices Meeting (IEDM)* (IEEE, 2015), pp. 16–17.
[69]K. Moon, A. Fumarola, S. Sidler, J. Jang, P. Narayanan, R. M. Shelby, G. W. Burr, and H. Hwang, "Bidirectional non-filamentary RRAM as an analog neuromorphic synapse, Part I: Al/Mo/$Pr_{0.7}Ca_{0.3}MnO_3$ material improvements and device measurements," IEEE J. Electron Devices Soc. **6**, 146–155 (2018).
[70]C. Cagli, D. Ielmini, F. Nardi, and A. L. Lacaita, "Evidence for threshold switching in the set process of NiO-based RRAM and physical modeling for set, reset, retention and disturb prediction," in *2008 IEEE International Electron Devices Meeting* (IEEE, 2008), pp. 1–4.
[71]C.-H. Cheng, C.-Y. Tsai, A. Chin, and F. S. Yeh, "High performance ultra-low energy RRAM with good retention and endurance," in *2010 International Electron Devices Meeting* (IEEE, 2010), pp. 14–19.
[72]K. Yang, J. Joshua Yang, R. Huang, and Y. Yang, "Nonlinearity in memristors for neuromorphic dynamic systems," Small Sci. **2**, 2100049 (2022).
[73]J. Tang, D. Bishop, S. Kim, M. Copel, T. Gokmen, T. Todorov, S. Shin, K.-T. Lee, P. Solomon, and K. Chan, "ECRAM as scalable synaptic cell for high-speed, low-power neuromorphic computing," in *2018 IEEE International Electron Devices Meeting (IEDM)* (IEEE, 2018), pp. 11–13.
[74]H. Liu, M. Wei, and Y. Chen, "Optimization of non-linear conductance modulation based on metal oxide memristors," Nanotechnol. Rev. **7**, 443–468 (2018).
[75]H.-M. Huang, Z. Wang, T. Wang, Y. Xiao, and X. Guo, "Artificial neural networks based on memristive devices: From device to system," Adv. Intell. Syst. **2**, 2000149 (2020).
[76]S. Sidler, I. Boybat, R. M. Shelby, P. Narayanan, J. Jang, A. Fumarola, K. Moon, Y. Leblebici, H. Hwang, and G. W. Burr, "Large-scale neural networks implemented with non-volatile memory as the synaptic weight element: Impact of conductance response," in *2016 46th European Solid-State Device Research Conference (ESSDERC)* (IEEE, 2016), pp. 440–443.
[77]S. Chandrasekaran, F. M. Simanjuntak, R. Saminathan, D. Panda, and T.-Y. Tseng, "Improving linearity by introducing Al in HfO_2 as a memristor synapse device," Nanotechnology **30**, 445205 (2019).
[78]Y. Jiang, K. Zhang, K. Hu, Y. Zhang, A. Liang, Z. Song, S. Song, and F. Wang, "Linearity improvement of HfO_x-based memristor with multilayer structure," Mater. Sci. Semicond. Process. **136**, 106131 (2021).
[79]S. Park, J. Noh, M.-l. Choo, A. M. Sheri, M. Chang, Y.-B. Kim, C. J. Kim, M. Jeon, B.-G. Lee, B. H. Lee, and H. Hwang, "Nanoscale RRAM-based synaptic electronics: Toward a neuromorphic computing device," Nanotechnology **24**, 384009 (2013).
[80]W. Wu, H. Wu, B. Gao, N. Deng, S. Yu, and H. Qian, "Improving analog switching in HfO_x-based resistive memory with a thermal enhanced layer," IEEE Electron Device Lett. **38**, 1019–1022 (2017).
[81]J. Woo, K. Moon, J. Song, S. Lee, M. Kwak, J. Park, and H. Hwang, "Improved synaptic behavior under identical pulses using AlO_x/HfO_2 bilayer RRAM array for neuromorphic systems," IEEE Electron Device Lett. **37**, 994–997 (2016).
[82]S. Kim, S. Choi, and W. Lu, "Comprehensive physical model of dynamic resistive switching in an oxide memristor," ACS Nano **8**, 2369–2376 (2014).
[83]B. Yan, D. Kuang, W. Wang, Y. Wang, B. Sun, and G. Zhou, "Investigation of multi-photoconductance state induced by light-sensitive defect in TiO_x-based memristor," Appl. Phys. Lett. **120**, 253506 (2022).
[84]J. Mao, L. Zhou, X. Zhu, Y. Zhou, and S. Han, "Photonic memristor for future computing: A perspective," Adv. Opt. Mater. **7**, 1900766 (2019).
[85]Q. Li, T. Wang, X. Hu, X. Wu, H. Zhu, L. Ji, Q. Sun, D. W. Zhang, and L. Chen, "Photonic memristor for future computing: A perspective," IEEE Electron Device Lett. **43**, 1089 (2022).
[86]X. Guan, W. Hu, M. A. Haque, N. Wei, Z. Liu, A. Chen, and T. Wu, "Light-responsive ion-redistribution-induced resistive switching in hybrid perovskite Schottky junctions," Adv. Funct. Mater. **28**, 1704665 (2018).
[87]H.-S. Jang, J.-H. Lee, Y.-S. Park, Y.-O. Kim, J. Park, T.-Y. Yang, K. Jin, J. Lee, S. Park, J. M. You, K. W. Jeong, A. Shin, I. S. Oh, M. K. Kwon, Y. I. Kim, H. H. Cho, H. N. Han, Y. Kim, Y. H. Chang, S. R. Paik, K. T. Nam, and Y. S. Lee, "Tyrosine-mediated two-dimensional peptide assembly and its role as a bio-inspired catalytic scaffold," Nat. Commun. **5**, 3665 (2014).
[88]M. Ju, O. H. Cho, J. Lee, S. D. Namgung, M.-K. Song, M. Balamurugan, J.-Y. Kwon, and K. T. Nam, "Quantitative analysis of the coupling between proton and electron transport in peptide/manganese oxide hybrid films," Phys. Chem. Chem. Phys. **22**, 7537–7545 (2020).
[89]J. Lee, I. R. Choe, Y. Kim, S. D. Namgung, K. Jin, H. Ahn, T. Sung, J. Kwon, Y. Lee, and K. T. Nam, "Proton conduction in a tyrosine-rich peptide/manganese oxide hybrid nanofilm," Adv. Funct. Mater. **27**, 1702185 (2017).
[90]Y. Wang, Z. Lv, Q. Liao, H. Shan, J. Chen, Y. Zhou, L. Zhou, X. Chen, V. A. L. Roy, Z. Wang, Z. Xu, Y. J. Zeng, and S. T. Han, "Synergies of electrochemical metallization and valance change in all-inorganic perovskite quantum dots for resistive switching," Adv. Mater. **30**, 1800327 (2018).
[91]M.-K. Song, S. D. Namgung, H. Lee, J. H. Yoon, Y.-W. Song, K. H. Cho, Y.-S. Lee, J.-S. Lee, K. T. Nam, and J.-Y. Kwon, "Tyrosine-mediated analog resistive switching for artificial neural networks," Nano Res. **16**, 858–864 (2022).
[92]F. H. Westheimer, "The magnitude of the primary kinetic isotope effect for compounds of hydrogen and deuterium," Chem. Rev. **61**, 265–273 (1961).
[93]H. Morgan, R. Pethig, and G. T. Stevens, "A proton-injecting technique for the measurement of hydration-dependent protonic conductivity," J. Phys. E: Sci. Instrum. **19**, 80 (1986).
[94]J. Wunsche, Y. Deng, P. Kumar, E. Di Mauro, E. Josberger, J. Sayago, A. Pezzella, F. Soavi, F. Cicoira, M. Rolandi, and C. Santato, "Protonic and electronic transport in hydrated thin films of the pigment eumelanin," Chem. Mater. **27**, 436–442 (2015).
[95]D. D. Ordinario, L. Phan, W. G. Walkup IV, J.-M. Jocson, E. Karshalev, N. Hüsken, and A. A. Gorodetsky, "Bulk protonic conductivity in a cephalopod structural protein," Nat. Chem. **6**, 596–602 (2014).
[96]G. Ding, B. Yang, R.-S. Chen, K. Zhou, S.-T. Han, and Y. Zhou, "MXenes for memristive and tactile sensory systems," Appl. Phys. Rev. **8**, 11316 (2021).
[97]M.-K. Song, Y.-W. Song, T. Sung, S. D. Namgung, J. H. Yoon, Y.-S. Lee, K. T. Nam, and J.-Y. Kwon, "Synaptic transistors based on a tyrosine-rich peptide for neuromorphic computing," RSC Adv. **11**, 39619–39624 (2021).
[98]X. Zhao, H. Xu, Z. Wang, Y. Lin, and Y. Liu, "Memristors with organic–inorganic halide perovskites," InfoMat **1**, 183–210 (2019).
[99]Z. Xiao, Y. Yuan, Y. Shao, Q. Wang, Q. Dong, C. Bi, P. Sharma, A. Gruverman, and J. Huang, "Giant switchable photovoltaic effect in organometal trihalide perovskite devices," Nat. Mater. **14**, 193–198 (2015).
[100]S. Lee, J. Sohn, Z. Jiang, H.-Y. Chen, and H.-S. Philip Wong, "Metal oxide-resistive memory using graphene-edge electrodes," Nat. Commun. **6**, 8407 (2015).
[101]A. N. Matsukatova, A. Y. Vdovichenko, T. D. Patsaev, P. A. Forsh, P. K. Kashkarov, V. A. Demin, and A. V. Emelyanov, "Scalable nanocomposite parylene-based memristors: Multifilamentary resistive switching and neuromorphic applications," Nano Res. **16**, 3207–3214 (2022).
[102]K. Yan, M. Peng, X. Yu, X. Cai, S. Chen, H. Hu, B. Chen, X. Gao, B. Dong, and D. Zou, "High-performance perovskite memristor based on methyl ammonium lead halides," J. Mater. Chem. C **4**, 1375–1381 (2016).
[103]J. H. Sung, J. H. Park, D. S. Jeon, D. Kim, M. J. Yu, A. C. Khot, T. D. Dongale, and T. G. Kim, "Retention enhancement through capacitance-dependent voltage division analysis in 3D stackable TaO_x/HfO_2-based selectorless memristor," Mater. Des. **207**, 109845 (2021).
[104]X. Zhao, S. Liu, J. Niu, L. Liao, Q. Liu, X. Xiao, H. Lv, S. Long, W. Banerjee, W. Li, S. Si, and M. Liu, "Confining cation injection to enhance CBRAM performance by nanopore graphene layer," Small **13**, 1603948 (2017).
[105]K. M. Kim, J. J. Yang, J. P. Strachan, E. M. Grafals, N. Ge, N. D. Melendez, Z. Li, and R. S. Williams, "Voltage divider effect for the improvement of variability and endurance of TaO_x memristor," Sci. Rep. **6**, 20085 (2016).
[106]A. Kumar, M. Das, V. Garg, B. S. Sengar, M. T. Htay, S. Kumar, A. Kranti, and S. Mukherjee, "Forming-free high-endurance Al/ZnO/Al memristor fabricated by dual ion beam sputtering," Appl. Phys. Lett. **110**, 253509 (2017).
[107]Y.-S. Chen, H.-Y. Lee, P.-S. Chen, W.-H. Liu, S.-M. Wang, P.-Y. Gu, Y.-Y. Hsu, C.-H. Tsai, W.-S. Chen, F. Chen, M. J. Tsai, and C. Lien, "Robust high-resistance state and improved endurance of HfO_X resistive memory by suppression of current overshoot," IEEE Electron Device Lett. **32**, 1585–1587 (2011).
[108]W. Banerjee, S. Maikap, C.-S. Lai, Y.-Y. Chen, T.-C. Tien, H.-Y. Lee, W.-S. Chen, F. T. Chen, M.-J. Kao, M.-J. Tsai, and J. R. Yang, "Formation polarity dependent improved resistive switching memory characteristics using nanoscale (1.3 nm) core-shell IrO_x nano-dots," Nanoscale Res. Lett. **7**, 194 (2012).

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© Author(s) 2023

[109]R. Cao, S. Liu, Q. Liu, X. Zhao, W. Wang, X. Zhang, F. Wu, Q. Wu, Y. Wang, H. Lv, S. Long, and M. Liu, "Improvement of device reliability by introducing a BEOL-compatible TiN barrier layer in CBRAM," IEEE Electron Device Lett. **38**, 1371–1374 (2017).
[110]A. M. Rana, T. Akbar, M. Ismail, E. Ahmad, F. Hussain, I. Talib, M. Imran, K. Mehmood, K. Iqbal, and M. Y. Nadeem, "Endurance and cycle-to-cycle uniformity improvement in tri-layered CeO_2/Ti/CeO_2 resistive switching devices by changing top electrode material," Sci. Rep. **7**, 39539 (2017).
[111]S. Fujii, J. A. C. Incorvia, F. Yuan, S. Qin, F. Hui, Y. Shi, Y. Chai, M. Lanza, and H.-S. P. Wong, "Scaling the CBRAM switching layer diameter to 30 nm improves cycling endurance," IEEE Electron Device Lett. **39**, 23–26 (2018).
[112]W. Wu, H. Wu, B. Gao, P. Yao, X. Zhang, X. Peng, S. Yu, and H. Qian, "A methodology to improve linearity of analog RRAM for neuromorphic computing," in *2018 IEEE Symposium on VLSI Technology* (IEEE, 2018), pp. 103–104.
[113]Z. Wang, M. Yin, T. Zhang, Y. Cai, Y. Wang, Y. Yang, and R. Huang, "Engineering incremental resistive switching in TaO_x based memristors for brain-inspired computing," Nanoscale **8**, 14015–14022 (2016).
[114]X. Zhao, Z. Wang, Y. Xie, H. Xu, J. Zhu, X. Zhang, W. Liu, G. Yang, J. Ma, and Y. Liu, "Photocatalytic reduction of graphene oxide-TiO_2 nanocomposites for improving resistive-switching memory behaviors," Small **14**, 1801325 (2018).

02 May 2026 09:56:35

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